On the Structure of Dynamical Laws

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Abstract

Dynamical theories are typically formulated by specifying a configuration space together with equations of motion. In general the equations are introduced independently of the geometric structure of the configuration space. However, compatibility geometry induces a rigidity linking representation invariants, admissible variational densities, differential order, and principal-symbol structure.

For the orthogonal compatibility configuration determined by a Riemannian manifold (M, h) and an orthogonal splitting $E = TM \oplus N$ with frozen tangent block, the compatible connections form a rigid affine configuration space. Local gauge-natural second-order dynamics on this configuration space possess a one-dimensional principal-symbol class generated by the quadratic curvature density. The associated Euler–Lagrange equation is

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

Introduction

Let (M, h) be a Riemannian manifold. If the metric h is fixed background data, the Levi–Civita theorem implies that metric-compatible connections on TM possess no independent degrees of freedom. Compatible connection dynamics therefore cannot occur on TM itself.

Nontrivial compatible connection variables require an extension of the tangent bundle. Consider an orthogonal bundle

$$E = TM \oplus N$$

equipped with the metric $g_E = h \oplus g_N$. Metric-compatible connections on E whose tangent block is fixed are necessarily of the form

$$\nabla^{\oplus} = \nabla^h \oplus \nabla^{\perp} + \Psi,$$

where $\nabla^{\perp}$ is a metric connection on N and the mixed component is determined by a tensor

$$K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N).$$

The admissible configuration space is therefore the affine space of compatible connections parameterized by $(\nabla^{\perp}, K)$.

Local gauge-natural dynamics on this configuration space are constrained by the compatibility geometry. Representation invariants, admissible variational densities, differential order, and principal-symbol structure are linked by the configuration geometry. The admissible leading-order variational density lies in the quadratic curvature class, and the associated Euler–Lagrange equation is

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

Section 1 constructs the configuration space determined by orthogonal compatibility. Section 2 determines the invariant geometric structure on that configuration space. Section 3 classifies admissible variational densities. Section 4 derives the Euler operator, Section 5 records the structural identities forced by gauge symmetry, Section 6 analyzes the linearized operator and symbol complex, and Section 7 states the final uniqueness results for admissible dynamics.

Technical derivations appear in Appendices A–D: Appendix A records the covariant variational structure, Appendix B derives the Euler–Lagrange equation, Appendix C analyzes the linearized symbol complex, and Appendix D establishes the Noether identities and curvature module.

In this program the configuration geometry is fixed first, and admissible local covariant scalar densities are then classified relative to that geometry.

Operator Convention. With the adjoint convention fixed by the L^2 pairing and Hodge metric, the Yang–Mills Euler–Lagrange equation is written equivalently as

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0 \qquad \text{and} \qquad (D^{\oplus})^{\dagger}\mathcal{F}^{\oplus} = 0.$$

Equivalently, the second form is the formal-adjoint rewriting of the first under this fixed convention. This should be distinguished from the Noether identity

$$(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} = 0,$$

where $\mathcal{E}_{\mathrm{YM}} = - \star D^{\oplus}(\star\mathcal{F}^{\oplus})$ is the Euler–Lagrange operator.

1 Configuration Geometry

1.1 Orthogonal Splitting

Definition 1.1 (Background Riemannian Geometry)**.** *Let (M, h) be a smooth n-dimensional Riemannian manifold with Levi–Civita connection ∇^h.*

Lemma 1.2 (Levi–Civita Rigidity)**.** *Let (M, h) be a Riemannian manifold. There exists a unique torsion-free connection on TM compatible with the metric h, namely the Levi–Civita connection.*

Proof. This is the fundamental theorem of Riemannian geometry; see [2, Vol. II]. □

Proposition 1.3 (Absence of Independent Compatible Connections on TM)**.** *If the metric h is fixed background data, then metric-compatible connections on TM possess no independent degrees of freedom.*

Proof. By Lemma 1.2, metric compatibility together with fixed h forces the tangent connection to be ∇^h. □

Proposition 1.4 (Necessity of Bundle Extension for Compatible Dynamics)**.** *If the tangent metric h is fixed, any compatible connection geometry with nontrivial degrees of freedom must be realized on an extension of the tangent bundle.*

Proof. Proposition 1.3 removes independent compatible connection variables on TM itself. Therefore nontrivial compatible connection dynamics requires an enlarged bundle carrying additional connection and mixed data. □

Definition 1.5 (Orthogonal Tangent–Normal Extension)**.** *Let $N \to M$ be a rank-k vector bundle with fiber metric g_N. Define the orthogonal extension*

$$E = TM \oplus N, \qquad g_E = h \oplus g_N.$$

1.2 Compatible Connections

Definition 1.6 (Orthogonal Compatibility Connection)**.** *An orthogonal compatibility connection on E is a metric connection of the block form*

$$\nabla^{\oplus} = \nabla^{h} \oplus \nabla^{\perp} + \Psi,$$

where $\nabla^{\perp}$ is a metric connection on N and

$$\Psi(X) := \begin{pmatrix} 0 & -K_X^* \\ K_X & 0 \end{pmatrix}$$

is determined by a mixed tensor $K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)$.

Definition 1.7 (Admissible Configuration Bundle)**.** *Let $\mathcal{C} \to M$ denote the affine bundle whose sections are orthogonal compatibility connections, equivalently pairs*

$$(\nabla^{\perp}, K) \in \Gamma\big(\mathcal{A}(N, g_N)\big) \times \Gamma(\mathrm{Sym}^2 T^*M \otimes N),$$

with frozen tangent block ∇^h.

Definition 1.8 (Configuration Space)**.** *The configuration space of admissible compatible connections is*

$$\mathcal{A}_{\oplus} := \Gamma(\mathcal{C}).$$

Remark 1.9 (Configuration Geometry)**.** *The configuration space arises from three structural inputs:*

- *fixed intrinsic geometry (M, h);*
- *orthogonal bundle extension $E = TM \oplus N$;*
- *compatibility constraint fixing the tangent block.*

Admissible connections are therefore parameterized by $(\nabla^{\perp}, K)$.

The remainder of this section analyzes the geometry of this configuration space, including its affine structure, gauge symmetry, and curvature decomposition.

Principal Bundle Formalism and Configuration Space. Only the configuration-space data required for the dynamical rigidity arguments are recorded here; the corresponding assembly, curvature-block, flatness-equivalence, and gauge-equivariance results are proved in Theorem A, Proposition 2.2, Theorem B, and Proposition 2.5 of [1]. The construction is expressed on the principal $\mathrm{SO}(E)$-bundle $P(E) \to M$ associated to (E, g_E), with principal connection $\omega^{\oplus}$, curvature $\mathcal{F}^{\oplus}$, and covariant exterior derivative $D^{\oplus}$. Under the standard correspondence between principal connections and associated covariant derivatives, the dynamical field is the metric-compatible covariant derivative $\nabla^{\oplus}$ on E. The principal variable $\omega^{\oplus}$ represents the admissible pair $(\nabla^{\perp}, K)$. Principal connections $\omega^{\oplus}$ and associated covariant derivatives $\nabla^{\oplus}$ are identified in the standard way. Notation: $\mathcal{F}^{\oplus}$ denotes the curvature of the admissible connection. Write $\mathrm{rank}(E) = m = n+k$ throughout.

The orthogonal compatibility ansatz restricts $\nabla^{\oplus}$ to the specific block form:

$$\nabla^{\oplus} = \nabla^{h} \oplus \nabla^{\perp} + \Psi, \qquad \Psi(X) := \begin{pmatrix} 0 & -K_X^* \\ K_X & 0 \end{pmatrix},$$

where ∇^h is the fixed Levi–Civita connection on TM, $\nabla^\perp$ is a dynamical metric connection on N, and the mixed block is first treated as

$$B \in \Omega^1\big(M, \mathrm{Hom}(TM, N)\big) \cong \Gamma(T^*M \otimes T^*M \otimes N).$$

The orthogonal-compatibility constraint is the symmetry condition

$$\Pi_{\mathrm{skew}}(B) = 0, \qquad \text{equivalently } B_{ij} = B_{ji}.$$

We write $K := \Pi_{\mathrm{sym}}(B) \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)$ for the admissible representative. For $X \in TM$, $K_X : TM \to N$ has adjoint $K_X^* : N \to TM$ defined by

$$\langle K_X(Y), \nu\rangle_{g_N} = h(Y, K_X^*(\nu)),$$

hence in an h-orthonormal frame (E_i):

$$K_X^*(\nu) = \sum_i \langle K(X, E_i), \nu\rangle_{g_N} E_i.$$

Definition 1.10 (Mixed-Derivative Operator on the Admissible Bundle)**.** *Let*

$$D^{h,\perp} : \Gamma(T^*M \otimes T^*M \otimes N) \to \Gamma(T^*M \otimes T^*M \otimes T^*M \otimes N)$$

denote the product covariant derivative induced by $(\nabla^h, \nabla^\perp)$*. In local coordinates:*

$$(D^{h,\perp}K)^\alpha_{\ell ij} = \partial_\ell K^\alpha_{ij} - \Gamma^r_{\ell i} K^\alpha_{rj} - \Gamma^r_{\ell j} K^\alpha_{ir} + (A_\ell)^\alpha{}_\beta\, K^\beta_{ij}.$$

Under admissibility $K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)$*, this operator is restricted to the symmetric mixed bundle and the notation* $D^{h,\perp}$ *is used uniformly throughout.*

1.3 Admissible Variables

The mixed variable and admissibility constraint are fixed as follows:

- unconstrained mixed variable: $B \in \Omega^1(M, \mathrm{Hom}(TM, N))$;
- admissible constraint: $\Pi_{\mathrm{skew}}(B) = 0$;
- admissible mixed tangent subspace:

$$V_{\mathrm{mix}} = \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \subset \Omega^1(M, \mathrm{Hom}(TM, N)).$$

All later projected operators on the mixed sector are restrictions of operators first defined on the B-model, then projected to V_{mix}.

Remark 1.11 (Concrete Fiber Model of the Admissible Configuration Bundle)**.** *The abstract bundle of Definition 1.7 is concretely realized as the affine subbundle of principal connections satisfying the orthogonal compatibility ansatz. Setting* $m = n + k$*, its model fiber is affine over*

$$\mathfrak{so}(k) \oplus \big(\mathrm{Sym}^2(\mathbb{R}^n)^* \otimes \mathbb{R}^k\big).$$

Thus the configuration space $\mathcal{A}_\oplus = \Gamma(\mathcal{C})$ *is the full space of admissible compatible connections with frozen tangent block.*

Dynamical Scope of the Admissible Class. Dynamics acts on the full space $\mathcal{A}_\oplus$: both $\nabla^\perp$ and K evolve. No $K \equiv 0$ truncation is imposed.

1.4 Curvature Decomposition

Proposition 1.12 (Gauss–Codazzi–Ricci Curvature Blocks)**.** *The curvature of an orthogonal compatibility connection decomposes into tangent, mixed, and normal blocks:*

$$\mathcal{F}^\oplus = \begin{pmatrix} R^h - K^* \wedge K & -(D^{h,\perp}K)^* \\ D^{h,\perp}K & F^\perp - K \wedge K^* \end{pmatrix}.$$

Thus the Gauss, Codazzi, and Ricci channels are the diagonal tangent block, the mixed derivative block, and the normal curvature block, respectively.

Proof. This is the block curvature formula for the orthogonal compatibility connection; see Proposition B.1 in Appendix B. □

2 Invariant Geometry of the Configuration Space

2.1 Gauge–Natural Structure

Let $O(M, h)$ and $O(N, g_N)$ denote orthonormal frame bundles. The structural isotropy group at a point is $O(n) \times O(k)$.

The vertical principal bundle automorphisms over id_M form the internal gauge group

$$\mathcal{G}_N \coloneqq \Gamma(\mathrm{SO}(N)).$$

Proposition 2.1 (Gauge Invariance of the Restricted Ansatz)**.** *The admissible configuration space $\mathcal{A}_\oplus$ is invariant under the restricted gauge group $\mathcal{G}_N$. Under natural inclusion $u \mapsto \mathrm{diag}(I_{TM}, u_N) \in \Gamma(\mathrm{SO}(E))$, the principal action $\omega^\oplus \mapsto u^{-1}\omega^\oplus u + u^{-1}du$ properly maps dynamically admissible connections to dynamically admissible connections.*

Proof. This is the block-restricted action inherited from the full $\mathrm{SO}(E)$ gauge action, with identity tangent block. Hence the frozen TT channel is preserved, normal components transform as a connection, and mixed components transform tensorially in $\Omega^1(\mathrm{Hom}(TM, N))$, preserving $\Pi_{\mathrm{skew}}(B) = 0$; compare Proposition 2.5 of [1]. □

Lemma 2.2 (Sobolev Smoothness of the Restricted Gauge Action)**.** *Let $s > n/2 + 1$. The map*

$$H^{s+1}(M, \mathrm{SO}(N)) \times \mathcal{A}^s_\oplus \to \mathcal{A}^s_\oplus, \qquad (u, \omega^\oplus) \mapsto u^{-1}\omega^\oplus u + u^{-1}du$$

with block embedding $\mathrm{diag}(I_{TM}, u)$ *is smooth.*

Proof. For $s > n/2 + 1$, $H^{s+1}(M, \mathrm{SO}(N))$ is a Hilbert Lie group; inversion and multiplication are smooth in this model. Sobolev multiplication is continuous for $s > n/2$, and $d : H^{s+1} \to H^s$ is continuous. Therefore $(u, \omega) \mapsto u^{-1}\omega u + u^{-1}du$ is smooth as a map into H^s-connections. The block embedding $\mathrm{SO}(N) \hookrightarrow \mathrm{SO}(E)$, $u \mapsto \mathrm{diag}(I_{TM}, u)$, is a smooth bundle morphism and preserves the frozen tangent block, so the action restricts smoothly to $\mathcal{A}^s_\oplus$. This is the standard Sobolev gauge-group smoothness mechanism in Yang–Mills settings; see [11, 14]. □

Definition 2.3 (Gauge-Natural Local Scalar Density)**.** *For a scalar density on $P(E) \to M$, the operator $\mathscr{L}$ on $\mathcal{C}$ is required to be natural under orthonormal frame changes and equivariant under internal gauge action $\mathcal{G}_N$.*

Let $\mathcal{F}_{\text{conf}}$ be the natural bundle functor associating (M, h) to the admissible configuration bundle $\mathcal{C} \to M$, and $\mathcal{F}_{\text{dens}}$ the functor associating it to the scalar density bundle $\Lambda^n T^*M$. Given the r-jet prolongation functor $J^r\mathcal{F}_{\text{conf}}$, an order-$r$ gauge-natural local scalar density *is a natural transformation $\mathscr{L}\colon J^r\mathcal{F}_{\text{conf}} \to \mathcal{F}_{\text{dens}}$ (see Kolář–Michor–Slovák, Ch. XII [3]).*

For any local morphism $\Phi = (\varphi, u) \in \mathcal{G}_{\text{loc}}$, naturality requires the following diagram to commute:

$$\begin{array}{ccc} J^r\mathcal{C} & \xrightarrow{\mathscr{L}} & \Lambda^n T^*M \\ \downarrow {\scriptstyle J^r\mathcal{F}_{\text{conf}}(\Phi)} & & \downarrow {\scriptstyle \mathcal{F}_{\text{dens}}(\varphi)} \\ J^r\mathcal{C} & \xrightarrow{\mathscr{L}} & \Lambda^n T^*M \end{array}$$

Equivariance is expressed on jet bundles by commutation with the induced action of $O(n) \times O(k)$ on the algebraic jet model at each point. Here $\mathcal{G}_{\text{loc}}$ denotes the local orthogonal-gauge groupoid acting on configuration jets.

Sobolev Setup. Let $s > n/2 + 1$ when functional-analytic statements are required. Then $\mathcal{A}^s_\oplus$ is the Hilbert completion of $\mathcal{A}_\oplus$, the restricted gauge group is $H^{s+1}(M, \mathrm{SO}(N))$, and Sobolev multiplication gives smooth gauge action on H^s-connections. Outside Appendices D–E, arguments are geometric and local and do not require Sobolev completion.

Definition 2.4 (Gauge-Natural Local Groupoid)**.** *Let $\mathcal{G}_{\text{loc}}$ be the groupoid of local morphisms*

$$\Phi = (\varphi, u),$$

where φ is an orthonormal frame morphism on TM and u is a local $\mathrm{SO}(N)$ gauge morphism on N. On local jets this acts by

$$(A, B, \partial A, \partial B) \mapsto (A^\Phi, B^\Phi, \partial A^\Phi, \partial B^\Phi),$$

where B^Φ is the induced representation action on $\Omega^1(M, \mathrm{Hom}(TM, N))$. The admissible subbundle is preserved:

$$\Pi_{\text{skew}}(B) = 0 \Longleftrightarrow \Pi_{\text{skew}}(B^\Phi) = 0.$$

Remark 2.5 (Global Assumptions and Proof Dependencies)**.** *A single global regime governs all subsequent integration arguments: either M is a compact manifold without boundary, or variations have compact interior support. This boundary condition applies universally.*

Furthermore, the structural proofs dynamically establish a strict hierarchy of geometric dependencies bounding explicit theorems:

- *The algebraic uniqueness of the symmetric scalar density (Theorem 2.13, Lemma 2.7, Theorem 3.37) depends on orthogonal structural-group symmetry and representation-induced pairing.*
- *The elimination of the independent $\mathfrak{so}(4)$ algebraic invariant (Theorem 2.10 together with Lemma 2.9) strictly requires representation-induced orthogonal functoriality in families.*
- *The isolation of the $m = 4$ topological Euler density (Theorem 3.37) requires continuous orientability of the full vector bundle ($w_1(E) = 0$).*

- *The symbol-level ellipticity modulo gauge of the local Yang–Mills operator (Theorem C.6) strictly depends on the stated linearization regime and compactness hypotheses for any global Fredholm interpretation.*

Unless otherwise specified, the local covariant constraint identities derived in Appendix B apply generically on arbitrary open subsets of the base manifold.

2.2 Invariant Pairings

Proposition 2.6. *$\mathcal{A}_\oplus$ is an affine space modeled on the vector subspace $V \subset \Omega^1(M, \mathfrak{so}(E))$, where vectors in V vanish on the $TM \to TM$ tangent block.*

Proof. Let $\nabla_0^\oplus \in \mathcal{A}_\oplus$. For any $\nabla^\oplus \in \mathcal{A}_\oplus$, set $a := \nabla^\oplus - \nabla_0^\oplus \in \Omega^1(M, \mathfrak{so}(E))$. Admissibility fixes the $TM \to TM$ block (frozen Levi–Civita block), while the normal variation $\delta\nabla^\perp$ and mixed variation δB remain. Imposing $\Pi_{\text{skew}}(\delta B) = 0$ gives the admissible mixed tangent space $V_{\text{mix}} = \Gamma(\text{Sym}^2 T^*M \otimes N)$. Hence a ranges over a linear subspace $V \subset \Omega^1(M, \mathfrak{so}(E))$, and $\mathcal{A}_\oplus = \nabla_0^\oplus + V$ is affine. □

2.3 Canonical Trace Pairing

For dynamic offsets $\dot\nabla_1, \dot\nabla_2 \in T_{\nabla^\oplus}\mathcal{A}_\oplus \cong V$, the canonical inner product is defined via global integration:

$$g_{L^2}(\dot\nabla_1, \dot\nabla_2) := \int_M \left\langle \dot\nabla_1, \dot\nabla_2 \right\rangle dV_h.$$

Lemma 2.7 (Invariant Bilinear Forms)**.** *Let $m = n + k$. The space of* Ad*-invariant symmetric bilinear forms $\text{Sym}^2(\mathfrak{so}(m)^*)^{\text{Ad}}$ is one-dimensional for $m \neq 4$, spanned uniquely by the trace form $B(X, Y) = \text{tr}(XY)$. For $m = 4$, the algebra decomposes as $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and the invariant space is two-dimensional, spanned by the independent Killing forms of the two ideals.*

Proof. For a simple Lie algebra $\mathfrak{g}$, the space of Ad-invariant symmetric bilinear forms is one-dimensional. Since $\mathfrak{so}(m)$ is compact simple for $m \geq 5$ and $m = 3$, any such form is a scalar multiple of the Cartan–Killing form $\kappa(X, Y) = (m - 2)\text{tr}(XY)$. Thus, $\dim(\text{Sym}^2(\mathfrak{so}(m)^*)^{\text{Ad}}) = 1$, uniquely spanned up to scale by the trace pairing $B(X, Y) = \text{tr}(XY)$.

For the exceptional case $m = 4$, the Lie algebra is semisimple $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. Schur's Lemma applies independently to the adjoint representation of each simple ideal. Therefore, any invariant symmetric bilinear form is a linear combination $a\kappa_L + b\kappa_R$, where $\kappa_{L,R}$ are the Killing forms of the respective $\mathfrak{su}(2)$ ideals, making the invariant space 2-dimensional; see Hall (2015), Chapter 10 [5]. □

Definition 2.8 (Representation-Induced Pairing)**.** *The invariant bilinear form on $\mathfrak{so}(E)$ is required to arise functorially from the defining orthogonal representation* $\text{SO}(E) \subset \text{End}(E)$*. Equivalently, it is natural with respect to orthogonal bundle isomorphisms $E \to E'$.*

Lemma 2.9 (Representation-Induced Restriction Principle)**.** *Let $V = \mathbb{R}^m$ with its Euclidean structure. Any symmetric bilinear form on $\mathfrak{so}(V)$ that is natural under orthogonal bundle isomorphisms and induced from the ambient endomorphism representation is the restriction of an $O(V)$-invariant symmetric bilinear form on* $\text{End}(V)$*. Hence it is a scalar multiple of* $\text{tr}(XY)$ *on $\mathfrak{so}(V)$.*

Proof. Naturality under orthogonal bundle isomorphisms gives conjugation invariance under $O(V)$. By the representation-induced hypothesis, the form is obtained by restricting an $O(V)$-invariant

symmetric bilinear form on $\mathrm{End}(V)$. Such forms are generated by $\mathrm{tr}(XY)$ and $\mathrm{tr}(X)\mathrm{tr}(Y)$. Restriction to $\mathfrak{so}(V)$ removes the trace-trace channel since $\mathrm{tr}(X) = 0$, leaving only a scalar multiple of $\mathrm{tr}(XY)$. □

Theorem 2.10 (Structural Invariant Pairing Classification)**.** *Let $E \to M$ be an orthogonal rank-m bundle with $m \geq 3$. In the representation-induced admissible class, the $O(E)$-invariant symmetric bilinear forms on $\mathfrak{so}(E)$ compatible with the defining orthogonal representation form a one-dimensional trace class:*

$$B_{\mathrm{adm}}(A, B) = c\,\mathrm{tr}(A \circ B), \qquad c \neq 0.$$

For $m = 4$, the algebraic semisimple freedom $a\kappa_L + b\kappa_R$ is removed by the representation-induced family-functorial condition, so only the trace-restriction class is admissible.

Proof. For $m \neq 4$, the Lie algebra $\mathfrak{so}(m)$ is simple and invariant symmetric bilinear forms are one-dimensional. For $m = 4$, write $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$, so invariant forms are $a\kappa_L + b\kappa_R$. By Lemma 2.9, only the representation-induced orthogonal trace class is functorial under orthogonal bundle isometries in families. Therefore the trace-induced class is unique up to scale in all dimensions. □

Proposition 2.11 (Trace-Class Pairing Uniqueness Up to Scale)**.** *Under the representation-induced requirement, the admissible invariant pairing on $\mathfrak{so}(E)$ is unique up to scale:*

$$B_{\mathrm{adm}}(A, B) = c\,\mathrm{tr}(A \circ B), \qquad c \neq 0.$$

Proof. This is immediate from Lemma 2.9 and Theorem 2.10: the representation-induced admissible class is one-dimensional and generated by the orthogonal trace class. □

Remark 2.12 (Pairing Normalization Convention)**.** *The multiplicative constant in Proposition 2.11 is a normalization convention fixed once in this paper (equivalently one may set $c = -1$ and use $-\mathrm{tr}$). This convention is not part of the uniqueness proof. The corresponding scale is global for the classified covariant density together with the induced configuration-space metric and is absorbed into the overall normalization parameter, not into the Yang–Mills coupling g alone.*

Theorem 2.13 (Canonical Kinetic Metric Uniqueness)**.** *Under the representation-induced pairing requirement with positive ultralocal mechanical pairing on admissible velocity fibers, on the admissible configuration space $\mathcal{A}^s_\oplus$, the induced ultralocal invariant kinetic metric obtained by restricting the trace pairing class on $\Omega^1(M, \mathfrak{so}(E))$ to the admissible tangent bundle is unique up to a global scale. After fixing the normalization convention of Remark 2.12, this yields the canonical L^2 metric defining $T(\dot\omega^\oplus) = \frac{1}{2}\|\dot\omega^\oplus\|^2_{L^2}$.*

Proof. At each $x \in M$, admissible tangent data decomposes as

$$V_x \cong \big(T^*_x M \otimes \mathfrak{so}(k)\big) \;\oplus\; \big(\mathrm{Sym}^2 T^*_x M \otimes N_x\big).$$

The first summand is the normal sector and the second is the admissible mixed sector. Under $O(n) \times O(k)$ these are non-isomorphic finite-dimensional representations, so any invariant bilinear cross-term between them vanishes by complete reducibility (compact isotropy) and Schur-type orthogonality.

Hence an ultralocal invariant fiber metric is determined by invariant pairings on each summand. In general this gives one scale per irreducible channel. The single-scale statement here is the admissibility condition that the kinetic form is the restriction of the ambient trace-induced Yang–Mills pairing on

$$\Omega^1(M, \mathfrak{so}(E))$$

to the admissible subbundle. Under this restriction, both sectors are measured by the same representation-induced trace class from Definition 2.8; by Lemma 2.7 and Proposition 2.11, this pairing is unique up to one global constant. Integrating over M yields exactly a global-scale multiple of the canonical L^2 metric, with normalization fixed only by convention. □

Proposition 2.14 (Invariant Pairing Forces the Velocity Hessian)**.** *For admissible mechanical systems on $\mathcal{A}_\oplus$, the velocity Hessian is forced by the same representation-induced invariant pairing that defines the curvature energy channel.*

Proof. An admissible mechanical Lagrangian has a fiberwise velocity Hessian on $T_\omega\mathcal{A}_\oplus$. Gauge-naturality forces this Hessian to be an invariant bilinear form on the admissible representation fibers. By Definition 2.8 and Proposition 2.11, the orthogonal representation induces a unique invariant trace class up to scale. Therefore the kinetic Hessian is fixed by that class, and the kinetic metric and curvature energy are generated by the same pairing. □

Define the geometric kinetic quantity

$$T(\dot{\nabla}^\oplus) := \frac{1}{2}\|\dot{\nabla}^\oplus\|^2_{L^2}.$$

3 Admissible Variational Densities

3.1 Locality and Differential Order

Remark 3.1 (Frozen TT-Block Scope)**.** *The Levi–Civita tangent block is fixed as part of the configuration object. Unfreezing the TT block enlarges the configuration space, introduces additional diffeomorphism-coupled symmetry content, and changes the classification problem; that regime is outside the present analysis. Freezing the Levi–Civita tangent block isolates compatibility-connection degrees of freedom and prevents mixing with metric variations; the enlarged problem including metric dynamics belongs to a different geometric regime.*

Definition 3.2 (Admissibility Regime)**.** *The admissibility regime on $\mathcal{A}_\oplus$ specifies the class of local gauge-natural covariant scalar densities whose structure is classified in this paper. A density belongs to this regime if it satisfies:*

1. *locality of the variational density;*
2. *gauge-naturality (orthonormal-frame naturality and restricted $\mathcal{G}_N$-equivariance);*
3. *semilinear principal symbol with field-independent principal coefficients at top order of the associated Euler–Lagrange operator.*

This regime defines the class of local gauge-natural covariant densities considered in the classification theorem below. All rigidity and uniqueness statements are relative to this regime; no claim is made for nonlocal, higher-jet, or non-semilinear densities.

Definition 3.3 (Admissible Mechanical System Regime)**.** *An admissible mechanical system on $\mathcal{A}_\oplus$ consists of a covariant scalar density in the admissibility regime of Definition 3.2 together with an ultralocal quadratic kinetic pairing on admissible velocity fibers.*

Remark 3.4 (Locality-to-Jet Reduction)**.** *In the gauge-natural framework of Kolář–Michor–Slovák, locality implies finite-jet locality via the Peetre–Slovák theorem; see Theorem 3.11.*

Remark 3.5 (Standing Nontrivial Gauge Regime)**.** *The classification analysis in this section is carried out in the nontrivial orthogonal-compatibility gauge regime*

$$n \geq 2, \qquad k \geq 2.$$

This is the natural geometric domain of the paper: $\Lambda^2 T^*M$ must be nontrivial for curvature-generated leading channels to exist, and $\mathfrak{so}(k)$ must be nontrivial for the normal gauge algebra to define a genuine compatibility gauge sector.

Remark 3.6 (Degeneracy at $n = 1$)**.** *When $n = 1$, one has $\Lambda^2 T^*M = 0$, hence the curvature two-form vanishes identically:*

$$\mathcal{F}^{\oplus} \equiv 0.$$

Accordingly, the curvature-based Yang–Mills classification problem collapses in this one-dimensional base regime.

Remark 3.7 (Degeneracy at $k = 1$)**.** *When $k = 1$, the normal gauge algebra is trivial:*

$$\mathfrak{so}(1) = 0.$$

The mixed tensor K remains, but the normal gauge-connection sector is absent. This is therefore a different geometric problem from the nontrivial compatibility gauge theory classified here.

Derived-Status Clarification. The properties appearing in the classification theorem are not independent axioms. The locality and differential-order structure follow from the covariant density admissibility regime through the derivation chain proved in Section 3 (Theorems 3.11, 3.12, 3.28), while the kinetic pairing rigidity enters only at the level of the separate mechanical system regime and follows from the representation-induced pairing classification in Section 2 (Theorem 2.13, Proposition 2.11).

Under Definition 3.2, the classification argument proceeds through the following steps:

finite-jet locality ⇒ first-jet top-order reduction ⇒ curvature-generated leading channel ⇒ quadratic invariant rig

Each step is established in the subsections below and culminates in Theorem 7.3.

Definition 3.8 (Orthogonal Compatibility Mechanical System)**.** *The orthogonal compatibility mechanical system is the triple*

$$(\mathcal{A}_{\oplus}, g_{L^2}, V),$$

where $\mathcal{A}_{\oplus}$ is the admissible affine configuration space, g_{L^2} is the canonical kinetic metric from Section 2, and

$$V(\omega^{\oplus}) = \frac{1}{2g^2} \int_M \left\langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \right\rangle \, dV_h$$

is the curvature potential.

3.2 Principal Symbol Constraints

Definition 3.9 (Semilinear Principal-Symbol Structure)**.** *Let $V \to M$ be the admissible variation bundle. A nonlinear Euler–Lagrange operator in this framework has semilinear principal-symbol structure if its principal symbol descends to a fixed $O(n) \times O(k)$-equivariant bundle morphism*

$$\sigma : S^*M \times V \to V$$

over S^*M, independent of configuration amplitude. In the structural regime, admissible principal symbols are forced to this semilinear form. In this paper, this is the normalized top-order notion: "semilinear principal-symbol structure" means semilinear principal symbol at top order.

Proposition 3.10 (Admissibility Criterion for Principal Type)**.** *In this paper, semilinear principal-symbol structure is the admissibility criterion at top order. Channels violating Definition 3.9 are excluded from the classified dynamics class.*

Proof. This is the operational criterion used in the exclusion propositions below (No-Go A/B and Proposition 3.32) and in the final classification theorem. □

Theorem 3.11 (Finite-Jet Locality)**.** *Under Definition 3.2, local gauge-natural scalar densities factor through finite jet bundles.*

Proof. This is the gauge-natural operator framework of Kolář–Michor–Slovák (Chapter XII), together with the Peetre–Slovák finite-order theorem in this setting [3]. The specific version used here is the finite-order locality theorem for gauge-natural operators (KMS Thm. 19.7 together with the natural-operator reduction in Chapter XII). □

Theorem 3.12 (First-Jet Top-Order Reduction)**.** *Under Definition 3.2, admissible top-order channels are forced to lie in the first-jet class of* $(\nabla^{\perp}, B)$ *in the orthogonal-compatibility model.*

Proof. Using Theorem 3.11 together with the order-counting result (Lemma 3.14), any top-order dependence beyond first jets would produce Euler–Lagrange channels that leave the admissible principal class by order/stability failure. Dependence on second or higher jets introduces curvature-derivative channels such as $D^{\oplus}\mathcal{F}^{\oplus}$, whose Euler–Lagrange operators are generically fourth order by Lemma 3.14. Hence higher-jet dependence contradicts the admissible semilinear principal-symbol class. □

Remark 3.13 (Working PDE Class)**.** *Under Definition 3.2, the working PDE class is semilinear at principal level and second-order in the covariant field channel; excluded channels are precisely those violating this principal-symbol criterion. This class corresponds to evolution laws whose principal propagation operator is background-determined and therefore admits a stable hyperbolic/elliptic symbol structure.*

Lemma 3.14 (Differential-Order Counting and Curvature Nonlinearity)**.** *For a local Lagrangian* $L = L(j^r\omega^{\oplus})$*, the Euler–Lagrange operator has order at most* $2r$*. In particular:*

- *if* $L = P(\mathcal{F}^{\oplus})$ *with no derivatives of* $\mathcal{F}^{\oplus}$*, then* $\mathcal{E}_L$ *is second-order in* $\omega^{\oplus}$*;*
- *if* L *depends on* $D^{\oplus}\mathcal{F}^{\oplus}$ *(or higher curvature jets), then* $\mathcal{E}_L$ *has order exceeding two in general and is excluded by admissibility.*

Nonlinear curvature monomials $P(\mathcal{F}^{\oplus})$ *of degree* ≥ 3 *are excluded by derived semilinear principal-symbol rigidity: their linearized principal symbols depend nontrivially on background amplitudes; see Proposition 3.17.*

Proof. The standard variational order bound gives $\mathrm{ord}(\mathcal{E}_L) \leq 2r$. Since $\mathcal{F}^{\oplus}$ is first-order in $\omega^{\oplus}$, curvature-only channels depend on first jets of $\omega^{\oplus}$, hence $r = 1$ and therefore second-order Euler–Lagrange operators. Dependence on $D^{\oplus}\mathcal{F}^{\oplus}$ raises jet order to $r = 2$ (or higher), so the Euler–Lagrange operator is generically of order ≥ 4 and violates admissibility. For nonlinear $P(\mathcal{F}^{\oplus})$, linearization produces principal coefficients built from $\mathcal{F}^{\oplus}(\omega_0)$, so the principal symbol varies with ω_0 and fails Definition 3.9. □

Theorem 3.15 (General Derivative-Curvature Jet-Order Exclusion)**.** *Let* L *be a local gauge-natural Lagrangian whose leading channel depends on curvature derivatives* $(D^{\oplus})^q\mathcal{F}^{\oplus}$ *with* $q \geq 1$*. Then the associated Euler–Lagrange operator has differential order strictly greater than two in* $\omega^{\oplus}$*. Therefore such channels are excluded from the admissible second-order class.*

Proof. Since $\mathcal{F}^{\oplus}$ is first-order in $\omega^{\oplus}$, $(D^{\oplus})^q\mathcal{F}^{\oplus}$ has jet order $q+1$. If the leading channel depends on $(D^{\oplus})^q\mathcal{F}^{\oplus}$ with $q \geq 1$, then the leading jet order satisfies $r \geq 2$. By variational order counting, the leading Euler channel has order at least $2r \geq 4$. Hence derivative-curvature leading channels lie outside the admissible semilinear second-order regime. □

Proposition 3.16 (No-Go A: Prototype Derivative Curvature Channel)**.** *For*

$$L_{D^{\oplus}\mathcal{F}^{\oplus}} = \|D^{\oplus}\mathcal{F}^{\oplus}\|^2_{L^2},$$

the Euler–Lagrange operator is fourth order in $\omega^{\oplus}$, as an immediate instance of Theorem 3.15.

Proof. This is the case $q = 1$ of Theorem 3.15. □

Proposition 3.17 (No-Go B: Higher Curvature Polynomial Channels)**.** *Let*

$$L_k(\omega^{\oplus}) = \operatorname{tr}((\mathcal{F}^{\oplus})^k), \qquad k \geq 3.$$

At background $\omega_0^{\oplus}$, one has

$$\delta L_k = k \operatorname{tr}(\delta\mathcal{F}^{\oplus}\,(\mathcal{F}_0^{\oplus})^{k-1}) + l.o.t., \qquad \delta\mathcal{F}^{\oplus} = D_0^{\oplus}(\delta\omega^{\oplus}) + l.o.t.$$

Hence the linearized principal symbol carries coefficients proportional to $(\mathcal{F}_0^{\oplus})^{k-2}$, so it depends nontrivially on background amplitude and violates Definition 3.9. Therefore higher curvature channels are excluded from admissible stable propagation.

Proof. Differentiate $L_k = \operatorname{tr}((\mathcal{F}^{\oplus})^k)$:

$$\delta \operatorname{tr}((\mathcal{F}^{\oplus})^k) = k \operatorname{tr}(\delta\mathcal{F}^{\oplus}\,(\mathcal{F}^{\oplus})^{k-1}).$$

At $\omega_0^{\oplus}$, substitute $\delta\mathcal{F}^{\oplus} = D_0^{\oplus}(\delta\omega^{\oplus}) + \text{l.o.t.}$ and integrate by parts to obtain the linearized Euler channel with top-order derivatives acting on $\delta\omega^{\oplus}$ and coefficient tensor built from $(\mathcal{F}_0^{\oplus})^{k-2}$. Therefore the principal symbol varies with $\omega_0^{\oplus}$ whenever $\mathcal{F}_0^{\oplus} \neq 0$, violating derived field-independent principal-symbol rigidity. □

Theorem 3.18 (Bifurcated Exclusion Mechanisms)**.** *In the admissibility regime, excluded non-Yang–Mills leading channels split into exactly two structural classes:*

(i) derivative-curvature channels fail by differential order (Theorem 3.15);

(ii) higher curvature polynomial channels fail by background-amplitude-dependent principal symbols (Proposition 3.17).

Proof. The derivative-curvature branch is Theorem 3.15. The higher-polynomial branch is Proposition 3.17. The first is order-theoretic; the second is semilinear principal-symbol failure. □

3.3 Curvature Forcing

Lemma 3.19 (Precise 1-Jet Model)**.** *Let $x \in M$. After choosing an h-orthonormal frame at x and a g_N-orthonormal frame of N_x, the fiber of the 1-jet bundle of admissible connections $J^1\mathcal{C}$ at x identifies with the affine space modeled on*

$$\big(T_x^*M \otimes \mathfrak{so}(k)\big) \;\oplus\; \big(T_x^*M \otimes T_x^*M \otimes N_x\big) \;\oplus\; \big(T_x^*M \otimes T_x^*M \otimes \mathfrak{so}(k)\big) \;\oplus\; \big(T_x^*M \otimes T_x^*M \otimes T_x^*M \otimes N_x\big)$$

with explicit coordinates corresponding to $(A_i, B_{ij}, \partial_\ell A_i, \partial_\ell B_{ij})$ in a normal chart. The admissible mixed subbundle is the projection $\Pi_{\text{skew}}(B) = 0$, i.e. $K_{ij} = B_{(ij)}$.

Proof. The statement follows from freezing the TT block, writing local admissible representatives as (A, B) in normal coordinates, and recording first derivatives at x. Full coordinate-level details are deferred to Appendix A. □

Lemma 3.20 (Gauge-Killing of Symmetric Normal-Connection Jets)**.** *For any point $x \in M$ and any symmetric tensor $S_{(\mu\nu)} \in \mathfrak{so}(k)$, there exists $u \in \mathcal{G}_N$ with $u(x) = I$ and $du(x) = 0$ such that, in any local trivialization,*

$$\partial_{(\mu}\widetilde{A}_{\nu)}(x) = \partial_{(\mu}A_{\nu)}(x) + S_{(\mu\nu)},$$

where A is the normal connection block. The transformed connection remains in $\mathcal{A}_\oplus$ and the frozen TT Levi–Civita block is unchanged.

Proof. Use the restricted block embedding $\widehat{u} = \mathrm{diag}(I_{TM}, u_N)$, so the frozen tangent block is preserved and the mixed block transforms tensorially. Choosing a local Lie-algebra jet with prescribed symmetric 2-jet and exponentiating produces the required shift of $\partial_{(\mu}A_{\nu)}(x)$. Full construction details are in Appendix A. □

Lemma 3.21 (Local Jet Surjectivity for Restricted Gauge Action)**.** *Fix $x \in M$. For any prescribed values*

$$u(x) = I, \qquad du(x) = 0, \qquad \partial_{(\mu}\partial_{\nu)}u(x) = S_{(\mu\nu)} \in \mathfrak{so}(k),$$

there exists a local $\mathrm{SO}(N)$ *gauge transformation realizing these jets.*

Proof. Apply Borel finite-jet realization to Lie-algebra valued jets and exponentiate to $\mathrm{SO}(N)$ [4, Thm. 1.2.6]. A cutoff equal to 1 near x preserves the prescribed jet and localizes the gauge representative. Details are in Appendix A. □

Lemma 3.22 (Frame Reduction on the Orthonormal Bundle)**.** *Fix $x \in M$. Under orthonormal-frame naturality and restricted gauge equivariance, an order-*1 *natural scalar density is determined by an $O(n) \times O(k)$-equivariant algebraic map on the 1-jet model at x.*

Proof. Using h-normal coordinates and a local orthonormal trivialization of N at x, the Levi–Civita block vanishes at x, so first jets reduce to algebraic tensors. Let $p \in O(M, h)_x$ and $q \in O(N, g_N)_x$. Naturality under frame changes sends jets in $J^1_x\mathcal{C}$ through the isotropy representation of $O(n)\times O(k)$. A natural scalar density is invariant under this isotropy action. Therefore it is encoded by an $O(n) \times O(k)$-equivariant map on the algebraic jet model. □

Proposition 3.23 (First-Jet Curvature Map and Complement Decomposition)**.** *Fix $x \in M$ and choose h-normal coordinates and an orthonormal normal frame at x. On the 1-jet fiber, define the linearized curvature map*

$$\mathscr{C}_x : \ (\partial A, \partial K) \longmapsto \Big((\partial_i A_j - \partial_j A_i),\ (\partial_\ell K_{ij} - \partial_i K_{\ell j})\Big),$$

with $K_{ij} = K_{ji}$. Then the jet-derivative space decomposes as

$$\partial A = \partial A^{\mathrm{curv}} \oplus \partial A^{\mathrm{sym}}, \qquad \partial K = \partial K^{\mathrm{curv}} \oplus \partial K^{\mathrm{sym}},$$

where

$$\partial A^{\mathrm{curv}}_{ij} := \partial_{[i}A_{j]}, \quad \partial A^{\mathrm{sym}}_{ij} := \partial_{(i}A_{j)},$$
$$\partial K^{\mathrm{curv}}_{\ell ij} := \partial_{[\ell}K_{i]j}, \quad \partial K^{\mathrm{sym}}_{\ell ij} := \partial_{(\ell}K_{i)j}.$$

The image of $\mathscr{C}_x$ is exactly $(\partial A^{\mathrm{curv}}, \partial K^{\mathrm{curv}})$, while the explicit complement is generated by $(\partial A^{\mathrm{sym}}, \partial K^{\mathrm{sym}})$.

Proof. The displayed map is the linearized first-jet part of the (NN) curvature and mixed curvature channels at x in normal coordinates, where background Christoffel symbols vanish at the base point. Antisymmetrization in (i, j) extracts $\partial A^{\mathrm{curv}}$ and antisymmetrization in (ℓ, i) extracts $\partial K^{\mathrm{curv}}$. The symmetric components in the same index pairs are annihilated by $\mathscr{C}_x$, and the direct-sum decomposition follows from the standard symmetric/antisymmetric splitting of rank-2 and rank-3 tensor slots. □

Theorem 3.24 (First-Jet Complement Exhaustion)**.** *Let Φ be an $O(n) \times O(k)$-equivariant first-order scalar invariant on the admissible 1-jet fiber. If Φ does not factor through the curvature map $\mathscr{C}_x$, then Φ depends on at least one complement generator from Proposition 3.23, namely $\partial A^{\mathrm{sym}}$ or $\partial K^{\mathrm{sym}}$.*

Proof. Proposition 3.23 gives a direct-sum splitting of derivative jets into curvature image plus complement. Therefore any invariant functional on the full 1-jet fiber decomposes as a function of these two sets of coordinates. If Φ were independent of $\partial A^{\mathrm{sym}}$ and $\partial K^{\mathrm{sym}}$, then Φ would factor through $\mathscr{C}_x$ by construction. Contraposition yields the claim. □

Proposition 3.25 (Non-Curvature First-Jet Channel)**.** *Under admissibility locality alone, order-1 scalar densities need not factor through $\mathcal{F}^{\oplus}$ alone. A local gauge-natural density with explicit ∇K dependence is*

$$\mathscr{L}_{\nabla K} := \|D^{h,\perp} K\|_h^2 \, dV_h.$$

Proof. $D^{h,\perp} K$ is tensorial, $O(n) \times O(k)$-equivariant, and gauge-natural. Its squared norm against (h, g_N) is a scalar density of first differential order in K. It is not determined by the pointwise value of $\mathcal{F}^{\oplus}$ alone because two 1-jets with equal $A, \partial A, K$ but different symmetric components of ∂K can have the same $\mathcal{F}^{\oplus}$ and different $\mathscr{L}_{\nabla K}$. This example shows that curvature reduction does not follow from locality alone; the principal-symbol admissibility condition is essential. □

Proposition 3.26 (Channel-by-Channel Exclusion Mechanisms)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, each first-jet complement channel from Proposition 3.23 is removed by a specific mechanism:*

- *$\partial A^{\mathrm{sym}}$ is gauge-removable by Lemma 3.20 and does not define an independent gauge-invariant leading channel.*
- *$\partial K^{\mathrm{sym}}$ yields mixed-derivative channels represented by $\|D^{h,\perp} K\|^2$, and Proposition 3.32 gives background-amplitude-dependent principal coefficients together with nonvanishing reduced projected obstruction (Theorem 3.31).*
- *derivative-curvature channels (for example $\|D^{\oplus} \mathcal{F}^{\oplus}\|^2$) are excluded by order via Theorem 3.15.*
- *higher curvature polynomial channels are excluded by background-dependent principal symbols via Proposition 3.17.*

Proof. The first item is Lemma 3.20. The second item is Proposition 3.32 together with Theorem 3.31 (valid for $n \geq 2,\ k \geq 2$). The third and fourth items are Theorem 3.15 and Proposition 3.17. Each mechanism contradicts Definition 3.9 or the second-order admissible class in Definition 3.2. Therefore no complement channel remains admissible at top order. □

Lemma 3.27 (Principal-Symbol Obstruction for Non-Curvature First-Jet Channels)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, let a first-jet admissible density have leading dependence on a non-curvature first-jet complement of the curvature map on $(A, B, \partial A, \partial B)$.*

Then its Euler linearization either exceeds second order or has background-amplitude-dependent top-order coefficients, hence violates Definition 3.2.

Proof. Proposition 3.23 gives the explicit first-jet curvature image and complementary channels. Theorem 3.24 reduces any non-curvature leading invariant to dependence on those complement generators. Proposition 3.26 removes each generator by explicit order/symbol mechanisms. Therefore any non-curvature first-jet leading dependence violates the admissible semilinear second-order class. □

Theorem 3.28 (Curvature-Generated Leading Channel)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, admissible leading-order scalar densities correspond to $O(n) \times O(k)$-invariant smooth functions on the first-jet model of Lemma 3.19, which by invariant theory factor through polynomial invariant generators; admissibility then forces the leading channel to factor through curvature invariants. Independent non-curvature first-jet leading channels are excluded.*

Proof. By Theorem 3.11, locality forces dependence through finite jets. By Theorem 3.12, admissible top-order channels reduce to first jets in this setting. Hence each admissible density is a smooth $O(n) \times O(k)$-invariant function on the finite-dimensional 1-jet fiber. Lemma 3.19 identifies the jet coordinates $(A, B, \partial A, \partial B)$, and Lemma 3.22 reduces classification to isotropy invariants; see also natural-operator reductions in [10, 3, 12]. Because $O(n) \times O(k)$ is compact, invariant theory applies and the algebra of polynomial invariants is generated by orthogonal contractions (Weyl invariant theory [7]). By Schwarz's theorem (Topology 14 (1975), Thm. 1), smooth invariant functions are smooth functions of polynomial generators [6]; see also [8] and Procesi (2007) [9].

Proposition 3.25 shows that ∂K contributions are allowed at the pure locality level. Theorem 3.24 identifies all non-curvature first-jet invariant generators through the explicit complement of Proposition 3.23. Proposition 3.26 removes each complement channel by a specific admissibility obstruction. Hence Lemma 3.27 applies and admissible top-order dynamics is curvature-generated. Hence every admissible leading-order invariant lies in the curvature-generated invariant algebra, and independent non-curvature first-jet channels cannot satisfy the admissibility regime simultaneously. For the Yang–Mills curvature channel, the principal symbol at any admissible background is the metric Laplace-type symbol determined by h (and fixed bundle pairing), hence independent of $\omega^{\oplus}$ throughout the admissible sector and consistent with Definition 3.9. □

Remark 3.29 (Semilinearity of the Yang–Mills Channel)**.** *The Yang–Mills curvature channel satisfies the semilinear principal-symbol condition because the linearized top-order operator is Laplace-type, with principal coefficients determined solely by the background metric h together with the fixed representation-induced pairing on $\mathfrak{so}(E)$.*

Lemma 3.30 (Mixed Principal-Symbol Formula on the Admissible Symmetric Bundle)**.** *Fix $x \in M$, background (A_0, K_0), and $\xi \in T_x^*M \setminus \{0\}$. For the mixed channel generated by*

$$\mathscr{L}_{\nabla K} = \|D^{h,\perp} K\|^2,$$

the principal mixed-to-normal symbol on admissible variations

$$\kappa \in \mathrm{Sym}^2 T_x^* M \otimes N_x$$

has the fiberwise form

$$\big(P_A(\xi, K_0)\kappa\big)_i^a = C^a{}_{b\alpha\beta}\, K_{0\,jr}^{\alpha}\, \xi^j\, \kappa_{ir}^{\beta},$$

for structure coefficients $C^a{}_{b\alpha\beta}$ determined by the $\mathfrak{so}(k)$-action on N_x.

Proof. Linearize the A-Euler channel

$$E_A = \mathrm{ad}^*_K(D^{h,\perp}K)$$

at (A_0, K_0). The top-order mixed term in κ is $\mathrm{ad}^*_{K_0}(D_0\kappa)$. At symbol level, replace D_0 by $\xi \otimes (\cdot)$, giving $\mathrm{ad}^*_{K_0}(\xi\otimes\kappa)$. Writing this bilinear contraction in local orthonormal frames yields the displayed index formula. Because κ is admissible, $\kappa^\beta_{ir} = \kappa^\beta_{ri}$, so the formula is defined on $\mathrm{Sym}^2 T^*_x M \otimes N_x$. □

Theorem 3.31 (Structural Reduced-Symbol Obstruction for the Mixed Channel)**.** *Assume $n \geq 2$ and $k \geq 2$. Fix $x \in M$ and $K_0(x) \neq 0$. Let $\xi \in T^*_x M \setminus \{0\}$ and define*

$$P_A(\xi, K_0)\kappa := \mathrm{ad}^*_{K_0}(\xi \otimes \kappa), \qquad \pi_{\mathrm{red},\xi} : T^*_x M \otimes \mathfrak{so}(k) \to V_{\mathrm{red},\xi},$$

where

$$V_{\mathrm{red},\xi} := \{a \in T^*_x M \otimes \mathfrak{so}(k) \,:\, \iota_{\xi^\sharp} a = 0\}.$$

The assignment

$$(\xi, K_0, \kappa) \longmapsto \pi_{\mathrm{red},\xi}\big(P_A(\xi, K_0)\kappa\big)$$

is fiberwise canonical and $O(T_xM) \times O(N_x)$-equivariant. Moreover, for every $K_0(x) \neq 0$, there exist $\xi \neq 0$ and admissible $\kappa \in \mathrm{Sym}^2 T^*_x M \otimes N_x$ such that

$$\pi_{\mathrm{red},\xi}\big(P_A(\xi, K_0)\kappa\big) \neq 0.$$

Proof. The map

$$\mathcal{P}_{x,\xi,K_0} : \mathrm{Sym}^2 T^*_x M \otimes N_x \to V_{\mathrm{red},\xi}, \qquad \mathcal{P}_{x,\xi,K_0}(\kappa) = \pi_{\mathrm{red},\xi}\big(P_A(\xi, K_0)\kappa\big)$$

is a composition of natural tensor contractions and orthogonal projection, hence $O(T_xM) \times O(N_x)$-equivariant. It therefore suffices to show nonvanishing at one explicit representative. Choose orthonormal coordinates $(e^1, \ldots, e^n)$ at x. Since $K_0(x) \neq 0$, there exist indices (j_0, r_0, α_0) with

$$K^{\alpha_0}_{0\,j_0 r_0} \neq 0.$$

Set $\xi = e^{j_0}$. By Lemma 3.30,

$$\big(P_A(\xi, K_0)\kappa\big)^a_i = C^a{}_{b\alpha\beta}\, K^\alpha_{0\,jr}\, \xi^j\, \kappa^\beta_{ir}.$$

Choose a transverse index $s \neq j_0$. Define an admissible symmetric variation by

$$\kappa^{\beta_0}_{sr_0} = \kappa^{\beta_0}_{r_0 s} = 1, \qquad \kappa^\beta_{ij} = 0 \text{ otherwise},$$

so $\kappa \in \mathrm{Sym}^2 T^*_x M \otimes N_x$ and its support is in a transverse slot s. Choose gauge indices (a_0, b, β_0) with

$$C^{a_0}{}_{b\alpha_0\beta_0} \neq 0$$

(possible because the $\mathfrak{so}(k)$ action on N_x is nontrivial). For these choices, the $i = s$ component satisfies

$$\big(P_A(\xi, K_0)\kappa\big)^{a_0}_s = C^{a_0}{}_{b\alpha_0\beta_0}\, K^{\alpha_0}_{0\,j_0 r_0}\, \kappa^{\beta_0}_{sr_0} \neq 0.$$

Since $s \neq j_0$, this component is transverse to $\xi^\sharp$. Therefore it survives orthogonal projection to $V_{\mathrm{red},\xi}$:

$$\pi_{\mathrm{red},\xi}\big(P_A(\xi, K_0)\kappa\big) \neq 0.$$

Hence $\mathcal{P}_{x,\xi,K_0} \neq 0$ for $K_0 \neq 0$. □

Proposition 3.32 (Explicit Principal-Symbol Rigidity Exclusion for $\|D^{h,\perp}K\|^2$ Channel)**.** *Assume $n \geq 2$ and $k \geq 2$. Consider*

$$\mathscr{L}_{\nabla K} = \|D^{h,\perp}K\|^2,$$

with coupled variables (A, K), where $D^{h,\perp}$ contains the normal connection A. The corresponding Euler channels are

$$E_K = (D^{h,\perp})^\dagger D^{h,\perp}K, \qquad E_A = \mathrm{ad}^*_K(D^{h,\perp}K).$$

Linearize around a background (A_0, K_0) with $K_0 \neq 0$, using admissible variations $\kappa \in \mathrm{Sym}^2 T^*M \otimes N$:

$$\delta E_A = \mathrm{ad}^*_{K_0}(D_0\kappa) + \mathrm{ad}^*_\kappa(D_0K_0) + l.o.t.$$

Hence the highest-order channel in the A-equation for κ has symbol

$$\sigma_\xi(\delta E_A) \supset \mathrm{ad}^*_{K_0}(\xi \otimes \kappa),$$

whose coefficient depends on K_0. Therefore the principal propagation channel is field-amplitude dependent and violates field-independent principal-symbol rigidity. Because the coefficient tensor in the principal symbol contains K_0, the highest-order operator is quasilinear rather than semilinear in the sense of Definition 3.9; this failure persists after gauge reduction by Theorem 3.31.

Proof. From $\mathscr{L}_{\nabla K}$, variation in K (with $\kappa \in \mathrm{Sym}^2 T^*M \otimes N$) gives the rough Laplace channel $(D^{h,\perp})^\dagger D^{h,\perp}K$. Variation in A enters through the connection part of $D^{h,\perp}$:

$$\delta_A(D^{h,\perp}K) = a \cdot K, \qquad \delta_K(D^{h,\perp}K) = D^{h,\perp}\kappa.$$

After integration by parts, the A-Euler channel is $\mathrm{ad}^*_K(D^{h,\perp}K)$. Linearizing at (A_0, K_0) gives the displayed formula. Taking principal symbol in the κ-channel yields $\mathrm{ad}^*_{K_0}(\xi \otimes \kappa)$, explicitly K_0-dependent. This is exactly the field-independent principal-symbol failure. By Lemma 3.30, the top-order mixed-to-normal contribution on admissible symmetric κ has component form

$$\big(P_A(\xi, K_0)\kappa\big)^a_i = C^a{}_{b\alpha\beta}\, K^\alpha_{0\,jr}\, \xi^j\, \kappa^\beta_{ir},$$

so the leading coefficient tensor depends on K_0. Hence the operator is quasilinear at top order in this channel, whereas the structural regime requires a field-independent (semilinear) principal symbol. For reduced-symbol verification, write the mixed/normal principal block as

$$\sigma_\xi(\delta E) \sim \begin{pmatrix} |\xi|^2_h \,\mathrm{Id}_A & P_A(\xi, K_0) \\ 0 & |\xi|^2_h \,\mathrm{Id}_K \end{pmatrix}, \qquad P_A(\xi, K_0)\kappa = \mathrm{ad}^*_{K_0}(\xi \otimes \kappa).$$

Restricted gauge has principal symbol $\sigma_\xi(d)\epsilon = (\xi \otimes \epsilon, 0)$, so gauge reduction removes only the longitudinal normal block in A. Theorem 3.31 gives explicit (ξ, κ) with nonzero transverse projection $\pi_{\mathrm{red},\xi} \circ P_A(\xi, K_0)\kappa$ in the regime $n \geq 2,\ k \geq 2$. Therefore the K_0-dependent top-order coefficient survives reduced-symbol quotienting. □

3.4 Quadratic Curvature Density

Theorem 3.33 (Algebraic Quadratic Invariant Space)**.** *Fix $x \in M$ and set $W_x = \Lambda^2 T^*_x M \otimes \mathfrak{so}(m)$. Under the isotropy action of $O(n) \times SO(m)$:*

$$\Big(\mathrm{Sym}^2(W^*_x)\Big)^{O(n)\times SO(m)} = \begin{cases} \mathrm{span}\{\mathcal{I}_{\mathrm{met}}\}, & n \neq 4, \\ \mathrm{span}\{\mathcal{I}_{\mathrm{met}}, \mathcal{I}_{\mathrm{top}}\}, & n = 4, \end{cases}$$

where

$$\mathcal{I}_{\mathrm{met}}(F, F) = \langle F, F\rangle_h, \qquad \mathcal{I}_{\mathrm{top}}(F, F) = \langle F, \star_h F\rangle_h.$$

Proof. Any quadratic scalar is the composition of:

(i) an $O(n)$-invariant contraction on two copies of $\Lambda^2 T_x^* M$, and

(ii) an $SO(m)$-invariant symmetric bilinear form on $\mathfrak{so}(m)$.

By Lemma 2.7 and Theorem 2.10, the Lie-algebra part is fixed up to scale by the trace pairing class. For the form-index part, $O(n)$-invariant bilinear forms on Λ^2 are generated by:

$$(\alpha, \beta) \mapsto \langle \alpha, \beta \rangle_h, \qquad (\alpha, \beta) \mapsto \langle \alpha, \star_h \beta \rangle_h.$$

The second exists only when $\star_h : \Lambda^2 \to \Lambda^{n-2}$ preserves degree, hence only for $n = 4$. Therefore the invariant quadratic space has dimension 1 for $n \neq 4$ and dimension 2 for $n = 4$, with the stated spanning set. Equivalently, the ε-contraction requires a fixed oriented volume form, and for $k = 2$ one has $\star_h^2 = (-1)^{2(n-2)} = +1$, so the degree-preserving topological channel exists only in dimension 4. □

Theorem 3.34 (Index-Level Classification of Quadratic Contractions)**.** *Let (e_i) be an h-orthonormal basis of $T_x M$, and write*

$$F = \frac{1}{2} F^a_{ij}\, e^i \wedge e^j \otimes T_a, \qquad F^a_{ij} = -F^a_{ji},$$

for an orthonormal basis (T_a) of $\mathfrak{so}(m)$ in a chosen representative of the trace pairing class. Any $O(n) \times SO(m)$-invariant quadratic scalar in F is a linear combination of:

$$Q_1(F) = \sum_{i,j,a} F^a_{ij} F^a_{ij}, \qquad Q_2(F) = \sum_{i,j,k,\ell,a} \varepsilon_{ijk\ell} F^a_{ij} F^a_{k\ell} \quad (n = 4).$$

No other independent quadratic contraction exists.

Proof. This is the index-level expansion of Theorem 3.33: orthogonal invariance reduces Lie-algebra contractions to δ_{ab} in the trace class and tangent-index contractions to δ-channels plus the $n = 4$ ε-channel. The explicit index algebra is provided in Appendix B. □

Definition 3.35 (Lagrangian Equivalence in the Derived Sector)**.** *Two admissible Lagrangians L_1, L_2 are equivalent, written $L_1 \sim L_2$, if*

$$L_2 = \alpha L_1 + d\Theta + c$$

for some nonzero constant $\alpha \in \mathbb{R}$, local $(n-1)$-form Θ, and constant $c \in \mathbb{R}$. Equivalently, they differ by global multiplicative scale, total divergence, and additive constant.

Definition 3.36 (Euler-Neutral Additions)**.** *A local addition to a Lagrangian is called* Euler-neutral *if it does not change the Euler–Lagrange operator in the boundary regime (closed manifold or compactly supported variations). In this paper, total divergences and topological density additions in dimension four are Euler-neutral.*

Theorem 3.37 (Quadratic Curvature Density Classification)**.** *Let M^n be an oriented Riemannian manifold, and $P \to M$ a principal* $\mathrm{SO}(m)$*-bundle with $m = n + k$. Under Definition 3.2, any admissible gauge-invariant local scalar density whose leading channel is quadratic in $\mathcal{F}^{\oplus}$ is equivalent (in the sense of Definition 3.35) to a linear combination of the invariant densities:*

- $\mathrm{tr}(\mathcal{F}^{\oplus} \wedge \star \mathcal{F}^{\oplus})$ *(exists for all dimensions n),*

- $\operatorname{tr}(\mathcal{F}^{\oplus} \wedge \mathcal{F}^{\oplus})$ *(exists independently only for dimension $n = 4$).*

In the special case $n = 4$, the form $\operatorname{tr}(\mathcal{F}^{\oplus} \wedge \mathcal{F}^{\oplus})$ acts strictly as a total divergence. Under the boundary regime, it contributes no terms to the Euler-Lagrange field equations. This is a uniqueness statement only within the admissible-local-mechanics regime modulo Definition 3.35.

Proof. By Theorem 3.33 and Theorem 3.34, quadratic local invariants are linear combinations of the metric contraction and, only in dimension $n = 4$, the topological wedge contraction. This is the standard orthogonal invariant-theory/Chern–Weil classification at quadratic order; see [7, 8, 2].

The metric contraction gives the Yang–Mills density $\operatorname{tr}(\mathcal{F}^{\oplus} \wedge \star\mathcal{F}^{\oplus})$. In $n = 4$, the second generator is $\operatorname{tr}(\mathcal{F}^{\oplus} \wedge \mathcal{F}^{\oplus})$. Equivalently, under a chosen normalization of the trace pairing class and Hodge metric, $\operatorname{tr}(\mathcal{F}^{\oplus} \wedge \star\mathcal{F}^{\oplus}) = \langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \rangle \, dV_h$.

For variational neutrality of the topological term, use $\delta\mathcal{F}^{\oplus} = D^{\oplus}(\delta\omega^{\oplus})$:

$$\delta \operatorname{tr}(\mathcal{F}^{\oplus} \wedge \mathcal{F}^{\oplus}) = 2 \operatorname{tr}(D^{\oplus}\delta\omega^{\oplus} \wedge \mathcal{F}^{\oplus}) = 2\, d \operatorname{tr}(\delta\omega^{\oplus} \wedge \mathcal{F}^{\oplus}) - 2 \operatorname{tr}(\delta\omega^{\oplus} \wedge D^{\oplus}\mathcal{F}^{\oplus}).$$

The Bianchi identity $D^{\oplus}\mathcal{F}^{\oplus} = 0$ removes the bulk term. Hence

$$\delta \int_M \operatorname{tr}(\mathcal{F}^{\oplus} \wedge \mathcal{F}^{\oplus}) = 2 \int_{\partial M} \operatorname{tr}(\delta\omega^{\oplus} \wedge \mathcal{F}^{\oplus}) = 0$$

under the boundary regime: either $\partial M = \varnothing$ (closed manifold) or variations are compactly supported, so boundary terms vanish. Therefore topological additions do not modify the Euler–Lagrange operator in the derived admissible sector. The exclusion mechanisms bifurcate as in Theorem 3.18: derivative-curvature channels are excluded by order (Theorem 3.15), while higher curvature polynomials are excluded by amplitude-dependent principal symbols (Proposition 3.17). □

4 Euler Operator

4.1 First Variation of the Curvature Energy

By Theorem 3.37, the admissible leading variational density is the quadratic curvature density. Appendix B computes its first variation in covariant form.

4.2 Divergence Form

Proposition 4.1 (Euler Operator in Divergence Form)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, the Euler operator of the classified curvature density is*

$$\mathcal{E}_{\mathrm{YM}} = - \star D^{\oplus}(\star\mathcal{F}^{\oplus}).$$

Proof. This is the covariant first-variation formula for the classified Yang–Mills density; see Appendix B and Proposition D.9. □

4.3 Yang–Mills Representation

Corollary 4.2 (Yang–Mills Representation)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, the admissible Euler–Lagrange equation is*

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

Proof. By Proposition 4.1, the Euler operator vanishes exactly when $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$. The explicit derivation is recorded in Appendix B. □

5 Structural Identities

5.1 Gauge Noether Identity

Corollary 5.1 (Gauge Noether Identity)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2$, $k \geq 2$, the gauge symmetry of the admissible Lagrangian yields the covariant Noether identity*

$$(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} = 0.$$

Proof. This is the variational consequence of gauge invariance for the classified Yang–Mills channel; see Theorem D.1 in Appendix D. □

5.2 Curvature–Module Containment

Corollary 5.2 (Curvature–Module Containment)**.** *For the classified Yang–Mills channel, the Euler operator belongs to the curvature-momentum-generated differential module:*

$$\mathcal{E}_{\mathrm{YM}} \in \mathcal{I}_{\mathrm{curv}}.$$

Proof. This is Theorem D.10 of Appendix D, specialized to the classified Yang–Mills curvature density. □

5.3 Differential Compatibility

The Noether identity and curvature-module containment together express the differential compatibility forced by the restricted gauge symmetry: the Euler operator is both gauge-compatible and curvature-momentum-generated throughout the admissible covariant regime.

6 Linearization Structure

6.1 Linearized Equation

Proposition 6.1 (Linearized Yang–Mills Equation)**.** *At a block-diagonal compatible background with $K_0 = 0$ and $F_0^{\perp} = 0$, linearization of the admissible Euler equation is governed at principal level by the Laplace-type operator*

$$\delta((D^{\oplus})^{\dagger}\mathcal{F}^{\oplus})\big|_{\omega_0^{\oplus}} \sim (D^{\oplus})_0^{\dagger} D_0^{\oplus}.$$

Proof. This is the linearization statement established in Appendix C. □

6.2 Principal Symbol

Corollary 6.2 (Principal Symbol)**.** *At the same background, the ungauged principal symbol of the linearized admissible equation is Laplace-type on the admissible variation bundle.*

Proof. This is Lemma C.5 in Appendix C. □

6.3 Symbol Complex

Corollary 6.3 (Gauge-Reduced Symbol Structure)**.** *In the regime $n \geq 2,\ k \geq 2$, linearization at a block-diagonal compatible background produces an operator that is elliptic modulo internal gauge, with the mixed sector fixed by the orthogonal-compatibility constraint.*

Proof. This follows from the linearization analysis, in particular Theorem C.6. The non-flat K_0-dependent mixed-symbol obstruction is excluded from the admissible principal class by Proposition 3.32 together with the structural reduced-symbol obstruction theorem 3.31, both in the regime $n \geq 2,\ k \geq 2$. The detailed symbol computation is recorded in Appendix C. □

7 Uniqueness of Admissible Dynamics

7.1 Admissible Euler Operators

Remark 7.1 (Single Pairing Controls Configuration Metric and Curvature Density)**.** *The same representation-induced* Ad*-invariant trace pairing on $\mathfrak{so}(E)$ generates both channels:*

- *the kinetic L^2 metric on $T\mathcal{A}_\oplus$;*
- *the quadratic curvature energy on $\mathcal{A}_\oplus$.*

Hence one algebraic pairing choice (up to scale) determines both the configuration-space metric and the classified covariant curvature density. Equivalently, the same geometric source determines curvature energy and kinetic metric in the admissible covariant regime.

Corollary 7.2 (Canonical Curvature Density)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, under Theorem 3.37, and in the normalization convention of Remark 2.12, the admissible covariant scalar density is uniquely determined up to Euler-neutral additions:*

$$\mathscr{L}_{\mathrm{YM}}(\omega^\oplus) := \frac{1}{2g^2} \left\langle \mathcal{F}^\oplus, \mathcal{F}^\oplus \right\rangle dV_h.$$

7.2 Classification Theorem

We now assemble the structural results of the preceding sections into the classification theorem for admissible local covariant scalar densities on the orthogonal-compatibility configuration space.

Theorem 7.3 (Main Rigidity Theorem (Admissible Dynamics Classification))**.** *Assume:*

1. *(M, h) is a smooth Riemannian manifold in the boundary regime (closed manifold or compactly supported variations), with $n = \dim M \geq 2$.*
2. *$E = TM \oplus N$ is an orthogonal splitting with fixed background metric data (h, g_N).*
3. *the normal rank satisfies $k = \mathrm{rank}(N) \geq 2$, so that the normal gauge algebra is nontrivial.*
4. *The tangent Levi–Civita block is frozen and the configuration space is the admissible affine space $\mathcal{A}_\oplus$.*
5. *The dynamics is given by local gauge-natural covariant scalar densities satisfying the admissibility regime of Definition 3.2, whose structural consequences, established in Sections 3–6, include finite-jet locality, first-jet reduction, Yang–Mills Euler representation, Noether compatibility, and gauge-reduced symbol structure.*

Under these conditions, and modulo the equivalence relation of Definition 3.35:

(i) every admissible gauge-natural local scalar density is equivalent, in the sense of Definition 3.35, to the Yang–Mills curvature density

$$\mathscr{L}_{\mathrm{YM}}(\omega^{\oplus}) = \frac{\lambda}{2g^2} \left\langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \right\rangle dV_h;$$

(ii) the Euler–Lagrange equation is

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0;$$

(iii) at principal-symbol level, modulo Euler-neutral additions, the admissible Euler–Lagrange principal-symbol class is one-dimensional and determined by the same representation-induced pairing;

Proof. The classification follows from the rigidity pipeline of Sections 3–6. Concretely, it combines finite-jet locality, curvature forcing, invariant rigidity, Euler-operator identification, Noether compatibility, symbol analysis, and the representation-induced pairing classification. Finite-jet locality is Theorem 3.11. First-jet top-order reduction is Theorem 3.12. Curvature forcing is Theorem 3.28 together with the structural exclusions (Proposition 3.32, Theorem 3.15, Proposition 3.17); the mixed reduced-symbol obstruction branch is used in the regime $n \geq 2,\ k \geq 2$. Quadratic invariant rigidity is Theorem 3.37. The normalization of the invariant trace class is fixed by Remark 2.12. Sections 4–6 identify the Euler operator, its Noether identity, and its gauge-reduced symbol class. Therefore all conclusions follow in the stated admissible class. □

7.3 Structural Consequences

Corollary 7.4 (Covariant Density Uniqueness)**.** *In the nontrivial orthogonal-compatibility gauge regime $n \geq 2,\ k \geq 2$, under Definition 3.2, and after fixing the normalization convention of Remark 2.12, the admissible covariant scalar density is unique modulo the equivalence relation of Definition 3.35:*

$$\mathscr{L}_{\mathrm{YM}}(\omega^{\oplus}) = \frac{1}{2g^2} \left\langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \right\rangle dV_h.$$

At principal-symbol level, modulo Euler-neutral additions, the associated Euler–Lagrange operator has a one-dimensional admissible class. All uniqueness here is relative to this structural regime.

Proof. This is the covariant-density form of Theorem 7.3, together with Corollary 7.2. □

A Covariant Variational Structure

Proposition A.1 (Explicit Yang–Mills First Variation and Potential Current)**.** *For the classified Yang–Mills density*

$$\mathrm{L}_{\mathrm{YM}} = \frac{1}{2} \left\langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \right\rangle dV_h,$$

the first variation is

$$\delta \mathrm{L}_{\mathrm{YM}} = \left\langle \delta\omega^{\oplus}, (D^{\oplus})^{\dagger} \mathcal{F}^{\oplus}, \right\rangle dV_h + d\Theta_L(\delta\omega^{\oplus}),$$

with

$$\Theta_L(\delta\omega^{\oplus}) = \left\langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle.$$

The sign convention is fixed by

$$\mathcal{E}_{\mathrm{YM}} = -\star D^{\oplus}(\star\mathcal{F}^{\oplus}).$$

Under this adjoint convention, $(D^{\oplus})^{\dagger}\mathcal{F}^{\oplus} = 0$ *is equivalent to* $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$.

Proof. Using $\delta\mathcal{F}^{\oplus} = D^{\oplus}(\delta\omega^{\oplus})$,

$$\delta \mathrm{L}_{\mathrm{YM}} = \left\langle \delta\mathcal{F}^{\oplus}, \star\mathcal{F}^{\oplus}, \right\rangle dV_h = \left\langle D^{\oplus}(\delta\omega^{\oplus}), \star\mathcal{F}^{\oplus}, \right\rangle dV_h.$$

Covariant integration by parts in the boundary regime fixed in the main text gives

$$\left\langle D^{\oplus}(\delta\omega^{\oplus}), \star\mathcal{F}^{\oplus}, \right\rangle dV_h = \left\langle \delta\omega^{\oplus}, (D^{\oplus})^{\dagger}\mathcal{F}^{\oplus}, \right\rangle dV_h + d\left\langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus}, . \right\rangle$$

Hence

$$\delta \mathrm{L}_{\mathrm{YM}} = \left\langle \delta\omega^{\oplus}, (D^{\oplus})^{\dagger}\mathcal{F}^{\oplus}, \right\rangle dV_h + d\Theta_L(\delta\omega^{\oplus}),$$

with $\Theta_L(\delta\omega^{\oplus}) = \langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus}\rangle$. □

Proposition A.2 (Covariant Presymplectic Current for the Yang–Mills Channel)**.** *The covariant presymplectic current is*

$$\omega_L(\delta_1\omega^{\oplus}, \delta_2\omega^{\oplus}) := \delta_1\Theta_L(\delta_2\omega^{\oplus}) - \delta_2\Theta_L(\delta_1\omega^{\oplus}),$$

and for the classified Yang–Mills channel this specializes to

$$\omega_L(\delta_1, \delta_2) = \delta_1 \left\langle \delta_2\omega^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle - \delta_2 \left\langle \delta_1\omega^{\oplus}, \star\mathcal{F}^{\oplus} \right\rangle.$$

Proof. This is the antisymmetrized field-space variation of the explicit potential current from the preceding proposition. Substituting $\Theta_L(\delta\omega^{\oplus}) = \langle \delta\omega^{\oplus}, \star\mathcal{F}^{\oplus}\rangle$ yields the displayed Yang–Mills specialization. □

Theorem A.3 (Gauge-Symmetry Noether Identity and Yang–Mills Specialization)**.** *For any smooth local gauge-invariant Lagrangian in this covariant framework, the Euler operator satisfies the symmetry identity*

$$(D^{\oplus})^{\dagger}E = 0.$$

For the classified Yang–Mills channel:

1. *the Euler–Lagrange equation is*

$$(D^{\oplus})^{\dagger}\mathcal{F}^{\oplus} = 0 \quad \Longleftrightarrow \quad D^{\oplus} \star \mathcal{F}^{\oplus} = 0;$$

2. *with Euler operator* $\mathcal{E}_{\mathrm{YM}} = -\star D^{\oplus}(\star\mathcal{F}^{\oplus})$*, restricted gauge symmetry* $\delta_\xi\omega^{\oplus} = D^{\oplus}\xi$ *yields the specialized covariant Noether identity*

$$(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} = 0.$$

Proof. Item (1) follows from the explicit first variation in Proposition 1 and arbitrariness of admissible variations. For the symmetry identity, gauge invariance gives

$$0 = \delta_\xi S = \int_M \left\langle E, D^{\oplus}\xi \right\rangle dV_h.$$

Covariant integration by parts yields

$$0 = -\int_M \left\langle (D^{\oplus})^{\dagger}E, \xi \right\rangle dV_h,$$

and arbitrariness of ξ implies $(D^{\oplus})^{\dagger}E = 0$. Applying this to $E = \mathcal{E}_{\mathrm{YM}}$ gives item (2). □

B Euler–Lagrange Derivation

This appendix derives the Euler–Lagrange operator associated with the classified Lagrangian of Theorem 7.3.

B.1 Explicit Block Curvature Formula

Proposition B.1 (Curvature Block Decomposition)**.** *With*

$$\omega^{\oplus} = \begin{pmatrix} \omega^h & -K^* \\ K & \omega^{\perp} \end{pmatrix},$$

the curvature splits as

$$\mathcal{F}^{\oplus} = \begin{pmatrix} R^h - K^* \wedge K & -(D^{h,\perp}K)^* \\ D^{h,\perp}K & F^{\perp} - K \wedge K^* \end{pmatrix},$$

where $D^{h,\perp}$ is the product covariant derivative induced by ω^h and $\omega^{\perp}$, acting on $T^*M \otimes T^*M \otimes N$ and restricting to $\mathrm{Sym}^2 T^*M \otimes N$ under the admissibility constraint $\Pi_{\mathrm{skew}}(K) = 0$.

Proof. Compute $\mathcal{F}^{\oplus} = d\omega^{\oplus} + \omega^{\oplus} \wedge \omega^{\oplus}$ blockwise. The mixed blocks arise from the covariant derivative $D^{h,\perp}K = dK + \omega^{\perp}K - K\omega^h$, while the diagonal blocks collect the background curvatures and quadratic mixed contractions. The adjoint placement follows from the convention for K_X^* introduced in Section 1. □

B.2 Covariant Yang–Mills Density

The classified admissible dynamics correspond to the quadratic curvature density

$$L(\omega^{\oplus}) = \tfrac{1}{2} \left\langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \right\rangle dV_h.$$

B.3 Euler–Lagrange Equation

For analytic statements, this equation is considered on the Sobolev configuration space $\mathcal{A}^s_{\oplus}$ with $s > n/2 + 1$.

Proposition B.2. *The first variation yields the Euler–Lagrange operator*

$$\mathcal{E}_{\mathrm{YM}} = - \star D^{\oplus}(\star \mathcal{F}^{\oplus}).$$

Proof. Using the identity $\delta\mathcal{F}^{\oplus} = D^{\oplus}(\delta\nabla^{\oplus})$ and the compatibility of the invariant pairing with $D^{\oplus}$, one has

$$\delta \int_M \tfrac{1}{2} \left\langle \mathcal{F}^{\oplus}, \mathcal{F}^{\oplus} \right\rangle dV_h = \int_M \left\langle D^{\oplus}\delta\nabla^{\oplus}, \mathcal{F}^{\oplus} \right\rangle dV_h.$$

Covariant integration by parts (using the boundary conditions fixed in the main text) gives

$$\int_M \left\langle \delta\nabla^{\oplus}, - \star D^{\oplus}(\star \mathcal{F}^{\oplus}) \right\rangle dV_h.$$

Stationarity for arbitrary admissible $\delta\nabla^{\oplus}$ gives the stated Euler form. □

B.4 Unified Curvature Dynamics

Theorem B.3 (Unified Equation)**.** *The Euler–Lagrange equation takes the unified Yang–Mills form*

$$D^{\oplus} \star \mathcal{F}^{\oplus} = 0.$$

Proof. By the proposition, $\mathcal{E}_{\mathrm{YM}} = 0$ is equivalent to $- \star D^{\oplus}(\star \mathcal{F}^{\oplus}) = 0$, hence $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$. □

B.5 Block Projection Equivalence

Proposition B.4.

$$\Pi_T(D^{\oplus} \star \mathcal{F}^{\oplus}) = 0, \quad \Pi_{\mathrm{mix}}(D^{\oplus} \star \mathcal{F}^{\oplus}) = 0, \quad \Pi_N(D^{\oplus} \star \mathcal{F}^{\oplus}) = 0$$

are equivalent to $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$.

Proof. Relative to $E = TM \oplus N$, $D^{\oplus} \star \mathcal{F}^{\oplus}$ is a 2×2 block matrix. Vanishing of all block components is equivalent to vanishing of the full endomorphism-valued form. □

Proposition B.5 (Explicit Block Form of $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$)**.** *Using Proposition B.1, the unified equation* $D^{\oplus} \star \mathcal{F}^{\oplus} = 0$ *is equivalent to the coupled block system*

$$\begin{aligned}
0 &= (D^{\oplus} \star \mathcal{F}^{\oplus})_{NN} = D^{\perp} \star \big(F^{\perp} - K \wedge K^*\big) \;+\; \mathcal{Q}_N(K, D^{h,\perp} K), \\
0 &= (D^{\oplus} \star \mathcal{F}^{\oplus})_{\mathrm{mix}} = D^{h,\perp} \star \big(D^{h,\perp} K\big) \;+\; \mathcal{Q}_{\mathrm{mix}}(K, F^{\perp}, R^h), \\
0 &= (D^{\oplus} \star \mathcal{F}^{\oplus})_{TT} = D^{h} \star \big(R^h - K^* \wedge K\big) \;+\; \mathcal{Q}_T(K, D^{h,\perp} K),
\end{aligned}$$

where each $\mathcal{Q}_\bullet$ *is an algebraic commutator–contraction expression determined by the block commutator action of* Ψ *on* $\star \mathcal{F}^{\oplus}$*, depending polynomially on* K *and its first derivatives.*

Proof. Insert the block decomposition of Proposition B.1 into $D^{\oplus} \star \mathcal{F}^{\oplus} = (D^{\mathrm{prod}} + [\Psi, \cdot]) \star \mathcal{F}^{\oplus}$, then project to (NN, mix, TT) blocks. The derivative parts give the displayed $D^{\perp}$, $D^{h,\perp}$, D^h channels; the remaining terms are exactly the algebraic commutator contractions collected as $\mathcal{Q}_\bullet$. □

C Linearization and Symbol Complex

C.1 Flat Vacuum

Proposition C.1 (Flat Vacuum)**.** $\mathcal{F}^{\oplus} = 0$ *is a solution of the Yang–Mills equation*

$$(D^{\oplus})^{\dagger} \mathcal{F}^{\oplus} = 0.$$

This links the flat solutions back to the flatness-equivalence result of Theorem B in [1]: vanishing assembled curvature is equivalent to vanishing of the Gauss, Codazzi, and Ricci compatibility blocks.

C.2 Linearization and Principal Symbol

Fix a background $\omega_0^{\oplus} = \omega^h \oplus \omega_0^{\perp} + \Psi(K_0)$ with

$$K_0 = 0, \qquad F(\omega_0^{\perp}) = 0.$$

Then $\omega_0^{\oplus}$ is block-diagonal and flat in the normal block.

In the orthogonal splitting $E = TM \oplus N$, admissible variations preserving the orthogonal splitting take the block form

$$\delta\omega^{\oplus} = \begin{pmatrix} 0 & -(\delta B)^* \\ \delta B & \delta\omega^{\perp} \end{pmatrix},$$

the curvature variation is

$$\delta\mathcal{F}^{\oplus} = D_0^{\oplus}(\delta\omega^{\oplus}),$$

and linearization of the Euler–Lagrange operator is

$$\delta\Big((D^{\oplus})^{\dagger}\mathcal{F}^{\oplus}\Big)\Big|_{\omega_0^{\oplus}} \;=\; (D^{\oplus})_0^{\dagger} D_0^{\oplus}(\delta\omega^{\oplus}),$$

at principal-symbol level, because the commutator contribution coming from the variation of the covariant derivative is lower order and therefore does not affect the principal symbol.

Lemma C.2 (Admissible Linearization at Block-Diagonal Compatible Background)**.** *At the above background, $\delta((D^{\oplus})^{\dagger}\mathcal{F}^{\oplus})$ preserves the admissible variation subbundle*

$$V \simeq \Omega^1(M, \mathfrak{so}(N)) \oplus \Omega^1(M, \mathrm{Hom}(TM, N)),$$

the admissible variation bundle.

Proof. Since the background connection preserves the orthogonal splitting $E = TM \oplus N$, the induced covariant derivative respects the normal and mixed bundles. The background operator $D_0^{\oplus}$ is block-diagonal and acts by induced covariant derivatives on these bundles. Therefore the linearized operator $(D^{\oplus})_0^{\dagger} D_0^{\oplus}$ maps each admissible block to admissible blocks without generating forbidden TT components. □

Lemma C.3 (Symmetry-Constraint Preservation)**.** *Let $\Pi_{\mathrm{sym}}, \Pi_{\mathrm{skew}}$ act on the mixed index pair of $B \in \Omega^1(\mathrm{Hom}(TM, N))$. At $K_0 = 0$ and $F_0^{\perp} = 0$, the mixed principal part commutes with these projectors, hence*

$$\Pi_{\mathrm{skew}}(\delta B_0) = 0 \implies \Pi_{\mathrm{skew}}(\delta B_t) = 0$$

for the linearized evolution.

Proof. At the block-diagonal compatible background with flat normal block, the mixed principal symbol coincides with the principal symbol of the rough Laplacian on the mixed bundle and acts componentwise on the two tangent-form indices. Therefore it commutes with symmetrization/skew-symmetrization in those indices, preserving the admissible symmetric subbundle. □

Lemma C.4 (Restricted Gauge Symbol in Admissible Blocks)**.** *For the restricted embedding $\widehat{u} = \mathrm{diag}(I_{TM}, u)$, the infinitesimal gauge action is*

$$\delta_\epsilon\omega^{\oplus} = D^{\oplus}\epsilon, \qquad \epsilon \in \Gamma(\mathfrak{so}(N)).$$

At a block-diagonal compatible background with $K_0 = 0$ and $F_0^{\perp} = 0$, its principal symbol is

$$\sigma_\xi(d)(\epsilon) = (\xi \otimes \epsilon, 0) \in \Omega^1(M, \mathfrak{so}(N)) \oplus \Omega^1(M, \mathrm{Hom}(TM, N)).$$

Here the first component lies in $\Omega^1(M, \mathfrak{so}(N))$. Hence gauge-image directions lie entirely in the normal block and have no mixed principal component.

Proof. With block-diagonal background connection, ϵ acts only in the normal block because the restricted gauge group acts only in the normal block. The principal part of $D^{\oplus}\epsilon$ is $d\epsilon$, giving $\xi \otimes \epsilon$ in the normal component. Mixed contributions involve the mixed compatibility block and therefore vanish at principal-symbol level when $K_0 = 0$. □

Lemma C.5 (Ungauged Principal Symbol)**.** *For $\xi \neq 0$, the ungauged principal symbol of $\delta((D^{\oplus})^{\dagger}\mathcal{F}^{\oplus})$ on*

$$V = \Omega^1(M, \mathfrak{so}(N)) \oplus \Omega^1(M, \mathrm{Hom}(TM, N))$$

is

$$\sigma_{\xi}^{\mathrm{ungauged}}(a) = |\xi|_h^2 a.$$

After gauge reduction in the normal block, the reduced symbol on the normal component is

$$\sigma_{\xi}^{\mathrm{red}}(a_{\perp}) = |\xi|_h^2 a_{\perp} - \xi \otimes \iota_{\xi^{\sharp}} a_{\perp}.$$

Gauge directions are normal longitudinal modes $a_{\perp} = \xi \otimes \phi$ with $\phi \in \mathfrak{so}(N)$, and these satisfy $\sigma_{\xi}^{\mathrm{red}}(a_{\perp}) = 0$.

Proof. The principal part of $(D^{\oplus})_0^{\dagger} D_0^{\oplus}$ is the Hodge Laplacian on bundle-valued 1-forms, giving $|\xi|^2$ times identity before gauge reduction. Passing to the gauge-reduced normal block yields the standard transverse expression; compare standard Yang–Mills linearization treatments in [12, 14, 15]. □

Theorem C.6 (Gauge-Reduced Ellipticity Statement)**.** *At the block-diagonal compatible background with $K_0 = 0$ and $F_0^{\perp} = 0$, the linearized operator*

$$\mathcal{L}: \; H^s\big(\Omega^1(M, \mathfrak{so}(N)) \oplus \Omega^1(M, \mathrm{Hom}(TM, N))\big) \to H^{s-2}\big(\Omega^1(M, \mathfrak{so}(N)) \oplus \Omega^1(M, \mathrm{Hom}(TM, N))\big)$$

has principal symbol

$$\sigma_{\xi}(\mathcal{L})(a_{\perp}, a_B) = \big(|\xi|_h^2 a_{\perp} - \xi \otimes \iota_{\xi^{\sharp}} a_{\perp},\; |\xi|_h^2 a_B\big), \qquad \xi \neq 0.$$

Its symbol kernel is exactly the longitudinal normal gauge directions $(\xi \otimes \phi, 0)$, $\phi \in \mathfrak{so}(N)$; the mixed sector has no nonzero symbol kernel and no longitudinal gauge modes because the restricted gauge group embeds trivially in the tangent block and acts only on the normal block. Therefore $\mathcal{L}$ is elliptic modulo internal gauge in the sense of principal-symbol complexes.

Projecting to the admissible symmetric mixed subbundle

$$V_{\mathrm{mix}} = \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \subset \Omega^1(M, \mathrm{Hom}(TM, N)),$$

the projected operator $\mathcal{L}_{\mathrm{adm}}$ has symbol

$$\sigma_{\xi}(\mathcal{L}_{\mathrm{adm}})(a_{\perp}, a_K) = \big(|\xi|_h^2 a_{\perp} - \xi \otimes \iota_{\xi^{\sharp}} a_{\perp},\; |\xi|_h^2 a_K\big).$$

At fixed $\xi \neq 0$, the symbol gauge map is

$$\sigma_{\xi}(d) : \mathfrak{g}_N \to V, \qquad \phi \mapsto (\xi \otimes \phi, 0).$$

The gauge-reduced symbol complex

$$0 \longrightarrow \mathfrak{g}_N \xrightarrow{\sigma_{\xi}(d)} V \xrightarrow{\sigma_{\xi}(\mathcal{L})} V_{\mathrm{red},\xi} \longrightarrow 0$$

is short exact, where $V_{\mathrm{red},\xi}$ is the ξ-transverse normal block plus full mixed block. Here the transverse normal block consists of normal 1-forms $a_{\perp}$ satisfying $\iota_{\xi^{\sharp}} a_{\perp} = 0$. This theorem establishes symbol-level ellipticity modulo gauge only. No global index or moduli-space dimension statement is asserted without additional compactness and boundary hypotheses.

Proof. The normal and mixed symbol formulas follow from Lemmas C.5, C.4, and C.3. Gauge-symbol image is longitudinal-normal only, so the mixed block has no gauge kernel and remains strictly elliptic. Exactness follows from $\sigma_\xi(\mathcal{L}) \circ \sigma_\xi(d) = 0$, together with the identification of the symbol kernel in Lemma C.5, and finite-dimensional dimension balance. Expanded symbol algebra and dimension counts are recorded in Appendix C. Since symbol fibers are finite-dimensional, injectivity together with the dimension balance yields exactness at the middle term. This completes the symbol-level ellipticity claim; no global Fredholm or moduli-dimension statement is asserted here. □

Proposition C.7 (Deformation-Complex Split at Block-Diagonal Compatible Background)**.** *At the block-diagonal compatible background with $K_0 = 0$ and $F_0^\perp = 0$, the principal-symbol deformation complex splits as:*

$$0 \longrightarrow \mathfrak{g}_N \xrightarrow{d} \Omega^1(M, \mathfrak{so}(N)) \xrightarrow{\mathcal{L}_N} \Omega^1(M, \mathfrak{so}(N)),$$

for the normal block, and

$$0 \longrightarrow \Omega^1(M, \mathrm{Hom}(TM, N)) \xrightarrow{\mathcal{L}_{\mathrm{mix}}} \Omega^1(M, \mathrm{Hom}(TM, N)),$$

for the mixed block.

Proof. The normal sequence is the standard Yang–Mills deformation symbol sequence from Theorem C.6. The mixed block receives no principal gauge image by Lemma C.4, and Theorem C.6 gives symbol $|\xi|_h^2 \mathrm{Id}$ on mixed variables. □

Corollary C.8 (Gauge-Rigid Mixed Sector)**.** *The mixed deformation block is strictly elliptic and contains no longitudinal gauge directions.*

Proof. By Proposition C.7, the mixed principal symbol is scalar Laplace-type and the restricted gauge image lies entirely in the normal block. □

Proposition C.9 (Constraint-Symbol vs Evolution-Symbol Compatibility)**.** *The symbol picture established by Theorem 4.5, Theorem 4.7, and Corollary 4.8 of [1] and the evolution symbol in this paper are compatible as the first-order deformation complex and the associated second-order Laplace-type evolution operator:*

- *in [1], the linearized constraint channel is first-order with wedge-type deformation-complex symbol;*
- *in this paper, linearization of $(D^\oplus)^\dagger \mathcal{F}^\oplus$ is second-order Laplace-type, obtained by adjoint composition of the deformation channel;*
- *under restricted gauge action, longitudinal gauge image appears only in the normal block, while the mixed block remains strictly elliptic.*

Proof. The first-order complex structure follows from the curvature/deformation framework of Theorem 4.5 and Theorem 4.7 of [1], with blockwise projection as in Corollary 4.8 there. Here the evolution linearization is $(D^\oplus)_0^\dagger D_0^\oplus$ at the block-diagonal compatible background with $K_0 = 0$ and $F_0^\perp = 0$, so its principal symbol is the Laplace-type composition of first-order symbols. Theorem C.6 and Proposition C.7 show that restricted gauge image is normal-longitudinal only, while the mixed block has scalar Laplace symbol and no gauge kernel. □

D Noether Identities and Curvature Module

D.1 Covariant Noether Identity

Theorem D.1. *For any smooth local gauge-invariant Lagrangian functional depending on the orthogonal connection $\omega^{\oplus}$, the Euler–Lagrange operator satisfies the covariant Noether identity*

$$(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} \equiv 0.$$

where $(D^{\oplus})^{\dagger}$ denotes the formal adjoint of $D^{\oplus}$ with respect to the pairing and the volume form.

Proof. For infinitesimal gauge parameter ϵ, $\delta_\epsilon \omega^{\oplus} = D^{\oplus}\epsilon$. Gauge invariance gives

$$0 = \delta_\epsilon S = \int_M \left\langle \mathcal{E}_{\mathrm{YM}}, D^{\oplus}\epsilon \right\rangle \, dV_h.$$

Here dV_h denotes the volume form induced by the base metric h. Also, $\langle \cdot, \cdot \rangle$ denotes the invariant trace pairing on $\mathrm{End}(E)$ induced by the orthogonal metric on E and the trace on endomorphisms. Using the compatibility of this pairing with the covariant derivative, covariant integration by parts gives

$$0 = -\int_M \left\langle (D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}}, \epsilon \right\rangle \, dV_h.$$

Arbitrariness of ϵ yields the identity. □

D.2 Orthogonal Channel Decomposition

Proposition D.2. *Represent the $\mathrm{End}(E)$-valued Euler operator $\mathcal{E}_{\mathrm{YM}}$ in the orthogonal splitting $E = TM \oplus N$ as*

$$\mathcal{E}_{\mathrm{YM}} = \begin{pmatrix} E_T & E_{\mathrm{mix}} \\ -(E_{\mathrm{mix}})^* & E_N \end{pmatrix}.$$

Then $(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} \equiv 0$ is equivalent to vanishing of each projected block component:

$$((D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}})_{TT} \equiv 0,$$
$$((D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}})_{TN} \equiv 0,$$
$$((D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}})_{NN} \equiv 0.$$

The off-diagonal couplings arise from the mixed compatibility block

$$\Psi \in \Omega^1\big(M, \mathrm{Hom}(TM, N) \oplus \mathrm{Hom}(N, TM)\big)$$

in the decomposition $D^{\oplus} = D^{\mathrm{prod}} + [\Psi, \cdot]$, and are fixed by the chosen adjoint convention.

Proof. The decomposition $E = TM \oplus N$ induces orthogonal projections associated with the splitting $E = TM \oplus N$ on $\mathrm{End}(E)$-valued forms. Applying these projections to the identity $(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} \equiv 0$ yields the three projected block identities. The coupling statement follows from splitting $D^{\oplus} = D^{\mathrm{prod}} + [\Psi, \cdot]$ and observing that $[\Psi, \cdot]$ maps diagonal and off-diagonal blocks into each other. □

Proposition D.3 (No-Extra-Constraints in the Gauge-Natural Symmetry Class)**.** *Within the symmetry class generated by internal gauge transformations $\mathcal{G}_N$ and orthonormal-frame naturality, the only universal off-shell differential identity in this symmetry class is $(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} \equiv 0$. Additional identities can occur only at special backgrounds with accidental stabilizers.*

Proof. Noether's second theorem applied to this local gauge-natural symmetry class yields $(D^{\oplus})^{\dagger}\mathcal{E}_{\mathrm{YM}} \equiv 0$. The statement is relative to this class: it does not exclude accidental identities at nongeneric backgrounds (for example flat or highly symmetric backgrounds), which are recorded as special cases rather than universal identities. □

D.3 Curvature-Ideal Closure

Structural role of $\mathcal{I}_{\mathrm{curv}}$. The module $\mathcal{I}_{\mathrm{curv}}$ is introduced to formalize that Euler channels in the admissible sector are generated from curvature momenta by iterated module operations and covariant differentiation/divergence. In particular, variation does not introduce an independent differential generator outside this curvature-momentum-generated module.

Definition D.4 (Curvature Momentum)**.** *For a smooth local gauge-invariant curvature Lagrangian $L = L(\mathcal{F}^{\oplus})$, define its curvature momentum by*

$$P_L := \frac{\partial L}{\partial \mathcal{F}^{\oplus}}.$$

For the quadratic Yang–Mills density,

$$L_{\mathrm{YM}} = \frac{1}{2}\langle \mathcal{F}^{\oplus}, \star\mathcal{F}^{\oplus}\rangle, \qquad P_{L_{\mathrm{YM}}} = \star\mathcal{F}^{\oplus}.$$

Definition D.5 (Algebraic Curvature-Momentum Module)**.** *Define*

$$\mathcal{J}_{\mathrm{curv}} := \left\{ \sum_i \alpha_i \wedge P_{L_i} \wedge \beta_i : \alpha_i, \beta_i \in \Omega^{\bullet}(M, \mathrm{End}(E)),\ L_i = L_i(\mathcal{F}^{\oplus}) \right\} \subset \Omega^{\bullet}(M, \mathrm{End}(E)).$$

This is the graded two-sided module over $\Omega^{\bullet}(M, \mathrm{End}(E))$ algebraically generated by curvature momenta under finite sums and module operations.

Definition D.6 (Differential Curvature-Momentum Module)**.** *Define $\mathcal{I}_{\mathrm{curv}}$ as the smallest graded two-sided module containing $\mathcal{J}_{\mathrm{curv}}$ and stable under both $D^{\oplus}$ and $(D^{\oplus})^{\dagger}$:*

$$\mathcal{I}_{\mathrm{curv}} \ = \bigcap \{\mathcal{M} \subset \Omega^{\bullet}(M, \mathrm{End}(E)) :\ \mathcal{J}_{\mathrm{curv}} \subset \mathcal{M},\ D^{\oplus}\mathcal{M} \subset \mathcal{M},\ (D^{\oplus})^{\dagger}\mathcal{M} \subset \mathcal{M}\}.$$

Lemma D.7 (Algebraic Closure of $\mathcal{J}_{\mathrm{curv}}$)**.** *The module $\mathcal{J}_{\mathrm{curv}}$ is closed under left/right wedge multiplication and the graded commutator.*

Proof. The wedge product and graded commutator are closed operations on $\Omega^{\bullet}(M, \mathrm{End}(E))$. Applying these operations to generators of the form $\alpha \wedge P_L \wedge \beta$ produces finite sums of elements of the same type, hence the module remains inside $\mathcal{J}_{\mathrm{curv}}$. □

Lemma D.8 (Differential Stability)**.** *The differential module $\mathcal{I}_{\mathrm{curv}}$ is stable under $D^{\oplus}$ and $(D^{\oplus})^{\dagger}$ by definition:*

$$D^{\oplus}(\mathcal{I}_{\mathrm{curv}}) \subseteq \mathcal{I}_{\mathrm{curv}}.$$
$$(D^{\oplus})^{\dagger}(\mathcal{I}_{\mathrm{curv}}) \subseteq \mathcal{I}_{\mathrm{curv}}.$$

Proof. Tautological from Definition D.6. □

Proposition D.9 (Euler Divergence Form for Curvature Lagrangians)**.** *For any smooth local gauge-invariant curvature Lagrangian $L = L(\mathcal{F}^{\oplus})$ with curvature momentum*

$$P := \frac{\partial L}{\partial \mathcal{F}^{\oplus}},$$

the Euler channel is

$$\mathcal{E}_{\mathrm{YM}} = -(D^{\oplus})^{\dagger} P.$$

For the quadratic Yang–Mills density $L = \frac{1}{2}\langle \mathcal{F}^{\oplus}, \star\mathcal{F}^{\oplus}\rangle$, one has $P = \star\mathcal{F}^{\oplus}$, hence

$$\mathcal{E}_{\mathrm{YM}} = -(D^{\oplus})^{\dagger}(\star\mathcal{F}^{\oplus}) = -\star D^{\oplus}(\star\mathcal{F}^{\oplus}).$$

Proof. For a curvature-dependent Lagrangian the Euler–Lagrange operator depends on the connection only through the variation of curvature. Using $\delta\mathcal{F}^{\oplus} = D^{\oplus}(\delta\omega^{\oplus})$ and assuming the usual vanishing boundary variations (or compact support of the variation):

$$\delta\int_M L(\mathcal{F}^{\oplus}) = \int_M \left\langle P, D^{\oplus}(\delta\omega^{\oplus})\right\rangle \;=\; -\int_M \left\langle (D^{\oplus})^{\dagger}P, \delta\omega^{\oplus}\right\rangle dV_h.$$

This is the standard Yang–Mills variational divergence form; compare [12, 14, 15]. Therefore the Euler channel is as stated. For $L = \frac{1}{2}\langle \mathcal{F}^{\oplus}, \star\mathcal{F}^{\oplus}\rangle$, one has $P = \star\mathcal{F}^{\oplus}$. □

Theorem D.10 (Euler Field Containment)**.** *For any admissible curvature Lagrangian, the Euler–Lagrange operator is the covariant divergence of the curvature momentum and belongs to the curvature-momentum-generated differential module:*

$$\mathcal{E}_{\mathrm{YM}} \in \mathcal{I}_{\mathrm{curv}}.$$

Proof. Let $P = \partial L/\partial\mathcal{F}^{\oplus}$ be the curvature momentum. By construction, $P \in \mathcal{J}_{\mathrm{curv}} \subseteq \mathcal{I}_{\mathrm{curv}}$. Proposition D.9 gives

$$\mathcal{E}_{\mathrm{YM}} = -(D^{\oplus})^{\dagger}P.$$

By Differential Stability, $(D^{\oplus})^{\dagger}P \in \mathcal{I}_{\mathrm{curv}}$. Hence $\mathcal{E}_{\mathrm{YM}} \in \mathcal{I}_{\mathrm{curv}}$. Thus admissible Euler channels remain confined to the curvature-momentum-generated module and introduce no independent differential generators beyond curvature within the admissible sector. □

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